

DC-DC Converters

Dc-dc converters are power electronic circuits that convert a dc voltage to a different dc voltage level, often providing a regulated output. The circuits described in this chapter are classified as switched-mode dc-dc converters, also called switching power supplies or switchers. This chapter describes some basic dc-dc converter circuits. Chapter 7 describes some common variations of these circuits that are used in many dc power supply designs.

6.1 LINEAR VOLTAGE REGULATORS

Before we discuss switched-mode converters, it is useful to review the motivation for an alternative to linear dc-dc converters that was introduced in Chapt. 1. One method of converting a dc voltage to a lower dc voltage is a simple circuit as shown in Fig. 6-1. The output voltage is

$$V_o = I_L R_L$$

where the load current is controlled by the transistor. By adjusting the transistor base current, the output voltage may be controlled over a range of 0 to roughly V_s . The base current can be adjusted to compensate for variations in the supply voltage or the load, thus regulating the output. This type of circuit is called a linear dc-dc converter or a linear regulator because the transistor operates in the linear region, rather than in the saturation or cutoff regions. The transistor in effect operates as a variable resistance.

While this may be a simple way of converting a dc supply voltage to a lower dc voltage and regulating the output, the low efficiency of this circuit is a serious drawback for power applications. The power absorbed by the load is $V_o I_L$, and

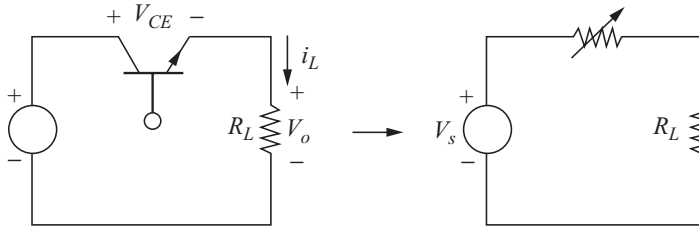


Figure 6-1 A basic linear regulator.

the power absorbed by the transistor is $V_{CE}I_L$, assuming a small base current. The power loss in the transistor makes this circuit inefficient. For example, if the output voltage is one-quarter of the input voltage, the load resistor absorbs one-quarter of the source power, which is an efficiency of 25 percent. The transistor absorbs the other 75 percent of the power supplied by the source. Lower output voltages result in even lower efficiencies. Therefore, the linear voltage regulator is suitable only for low-power applications.

6.2 A BASIC SWITCHING CONVERTER

An efficient alternative to the linear regulator is the switching converter. In a switching converter circuit, the transistor operates as an electronic switch by being completely on or completely off (saturation or cutoff for a BJT or the triode and cutoff regions of a MOSFET). This circuit is also known as a dc chopper.

Assuming the switch is ideal in Fig. 6-2, the output is the same as the input when the switch is closed, and the output is zero when the switch is open. Periodic

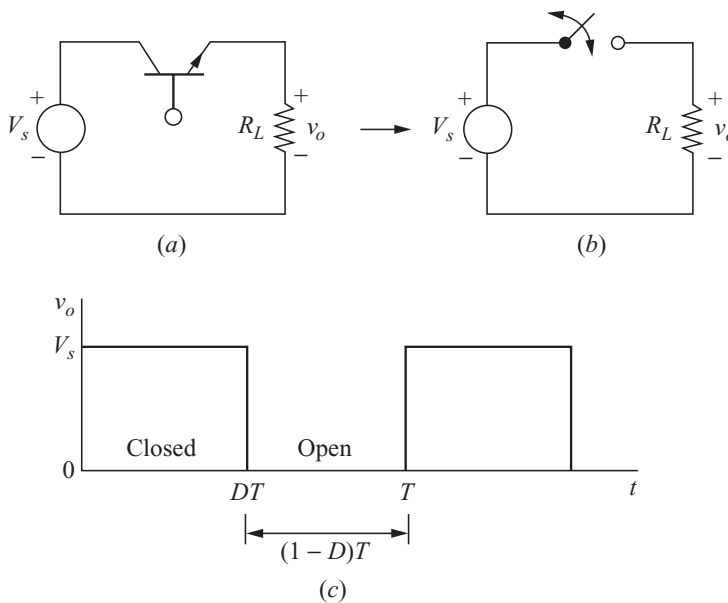


Figure 6-2 (a) A basic dc-dc switching converter; (b) Switching equivalent; (c) Output voltage.

opening and closing of the switch results in the pulse output shown in Fig. 6-2c. The average or dc component of the output voltage is

$$V_o = \frac{1}{T} \int_0^T v_o(t) dt = \frac{1}{T} \int_0^{DT} V_s dt = V_s D \quad (6-1)$$

The dc component of the output voltage is controlled by adjusting the duty ratio D , which is the fraction of the switching period that the switch is closed

$$D \equiv \frac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}} = \frac{t_{\text{on}}}{T} = t_{\text{on}} f \quad (6-2)$$

where f is the switching frequency. The dc component of the output voltage will be less than or equal to the input voltage for this circuit.

The power absorbed by the ideal switch is zero. When the switch is open, there is no current in it; when the switch is closed, there is no voltage across it. Therefore, all power is absorbed by the load, and the energy efficiency is 100 percent. Losses will occur in a real switch because the voltage across it will not be zero when it is on, and the switch must pass through the linear region when making a transition from one state to the other.

6.3 THE BUCK (STEP-DOWN) CONVERTER

Controlling the dc component of a pulsed output voltage of the type in Fig. 6-2c may be sufficient for some applications, such as controlling the speed of a dc motor, but often the objective is to produce an output that is purely dc. One way of obtaining a dc output from the circuit of Fig. 6-2a is to insert a low-pass filter after the switch. Figure 6-3a shows an LC low-pass filter added to the basic converter. The diode provides a path for the inductor current when the switch is opened and is reverse-biased when the switch is closed. This circuit is called a *buck converter* or a *step-down converter* because the output voltage is less than the input.

Voltage and Current Relationships

If the low-pass filter is ideal, the output voltage is the average of the input voltage to the filter. The input to the filter, v_x in Fig. 6-3a, is V_s when the switch is closed and is zero when the switch is open, provided that the inductor current remains positive, keeping the diode on. If the switch is closed periodically at a duty ratio D , the average voltage at the filter input is $V_s D$, as in Eq. (6-1).

This analysis assumes that the diode remains forward-biased for the entire time when the switch is open, implying that the inductor current remains positive. An inductor current that remains positive throughout the switching period is known as *continuous current*. Conversely, discontinuous current is characterized by the inductor current's returning to zero during each period.

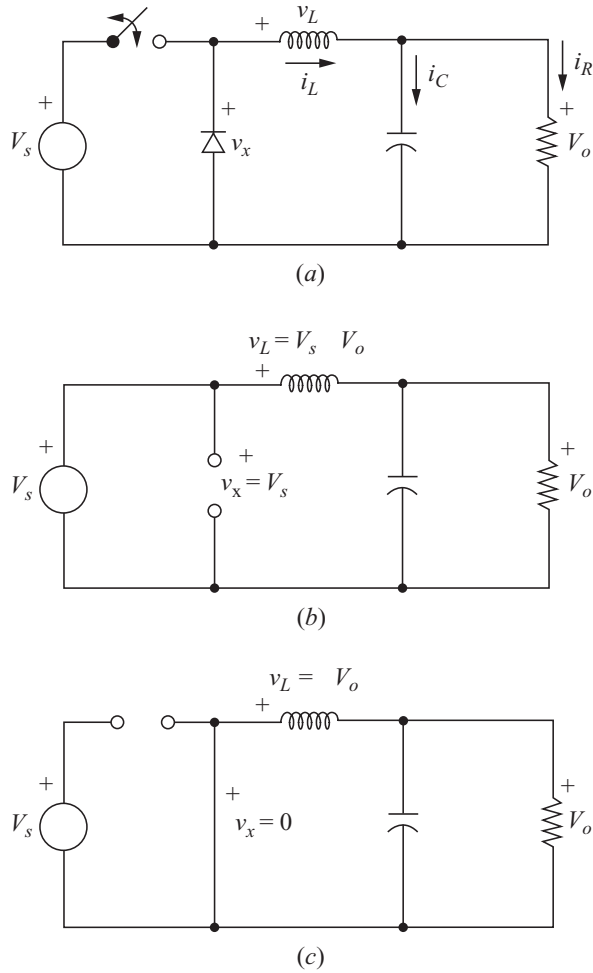


Figure 6-3 (a) Buck dc-dc converter; (b) Equivalent circuit for the switch closed; (c) Equivalent circuit for the switch open.

Another way of analyzing the operation of the buck converter of Fig. 6-3a is to examine the inductor voltage and current. This analysis method will prove useful for designing the filter and for analyzing circuits that are presented later in this chapter.

Buck converters and dc-dc converters in general, have the following properties when operating in the steady state:

1. The inductor current is periodic.

$$i_L(t + T) = i_L(t) \quad (6-3)$$

2. The average inductor voltage is zero (see Sec. 2.3).

$$V_L = \frac{1}{T} \int_t^{t+T} v_L(\lambda) d\lambda = 0 \quad (6-4)$$

3. The average capacitor current is zero (see Sec. 2.3).

$$I_C = \frac{1}{T} \int_t^{t+T} i_C(\lambda) d\lambda = 0 \quad (6-5)$$

4. The power supplied by the source is the same as the power delivered to the load. For nonideal components, the source also supplies the losses.

$$\begin{aligned} P_s &= P_o && \text{ideal} \\ P_s &= P_o + \text{losses} && \text{nonideal} \end{aligned} \quad (6-6)$$

Analysis of the buck converter of Fig. 6-3a begins by making these assumptions:

1. The circuit is operating in the steady state.
2. The inductor current is continuous (always positive).
3. The capacitor is very large, and the output voltage is held constant at voltage V_o . This restriction will be relaxed later to show the effects of finite capacitance.
4. The switching period is T ; the switch is closed for time DT and open for time $(1-D)T$.
5. The components are ideal.

The key to the analysis for determining the output V_o is to examine the inductor current and inductor voltage first for the switch closed and then for the switch open. The net change in inductor current over one period must be zero for steady-state operation. The average inductor voltage is zero.

Analysis for the Switch Closed When the switch is closed in the buck converter circuit of Fig. 6-3a, the diode is reverse-biased and Fig. 6-3b is an equivalent circuit. The voltage across the inductor is

$$v_L = V_s - V_o = L \frac{di_L}{dt}$$

Rearranging,

$$\frac{di_L}{dt} = \frac{V_s - V_o}{L} \quad \text{switch closed}$$

Since the derivative of the current is a positive constant, the current increases linearly as shown in Fig. 6-4b. The change in current while the switch is closed is computed by modifying the preceding equation.

$$\begin{aligned} \frac{di_L}{dt} &= \frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s - V_o}{L} \\ (\Delta i_L)_{\text{closed}} &= \left(\frac{V_s - V_o}{L} \right) DT \end{aligned} \quad (6-7)$$

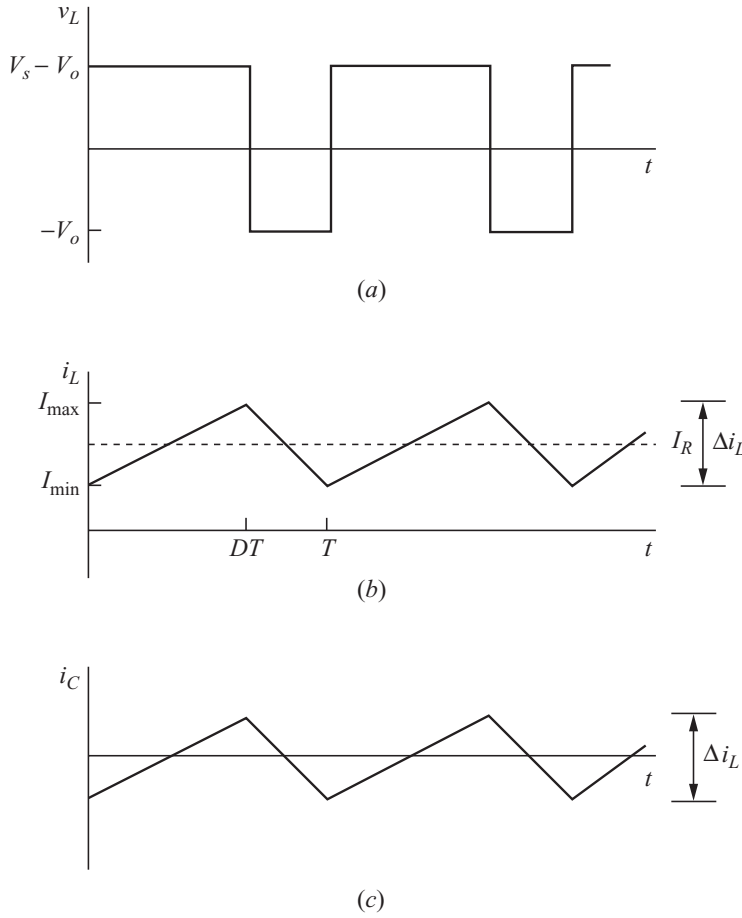


Figure 6-4 Buck converter waveforms: (a) Inductor voltage; (b) Inductor current; (c) Capacitor current.

Analysis for the Switch Open When the switch is open, the diode becomes forward-biased to carry the inductor current and the equivalent circuit of Fig. 6-3c applies. The voltage across the inductor when the switch is open is

$$v_L = -V_o = L \frac{di_L}{dt}$$

Rearranging,

$$\frac{di_L}{dt} = \frac{-V_o}{L} \quad \text{switch open}$$

The derivative of current in the inductor is a negative constant, and the current decreases linearly as shown in Fig. 6-4b. The change in inductor current when the switch is open is

$$\begin{aligned} \frac{\Delta i_L}{\Delta t} &= \frac{\Delta i_L}{(1-D)T} = -\frac{V_o}{L} \\ (\Delta i_L)_{\text{open}} &= -\left(\frac{V_o}{L}\right)(1-D)T \end{aligned} \quad (6-8)$$

Steady-state operation requires that the inductor current at the end of the switching cycle be the same as that at the beginning, meaning that the net change in inductor current over one period is zero. This requires

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

Using Eqs. (6-7) and (6-8),

$$\left(\frac{V_s - V_o}{L}\right)(DT) - \left(\frac{V_o}{L}\right)(1 - D)T = 0$$

Solving for V_o ,

$$\boxed{V_o = V_s D} \quad (6-9)$$

which is the same result as Eq. (6-1). *The buck converter produces an output voltage that is less than or equal to the input.*

An alternative derivation of the output voltage is based on the inductor voltage, as shown in Fig. 6-4a. Since the average inductor voltage is zero for periodic operation,

$$V_L = (V_s - V_o)DT + (-V_o)(1 - D)T = 0$$

Solving the preceding equation for V_o yields the same result as Eq. (6-9), $V_o = V_s D$.

Note that the output voltage depends on only the input and the duty ratio D . If the input voltage fluctuates, the output voltage can be regulated by adjusting the duty ratio appropriately. A feedback loop is required to sample the output voltage, compare it to a reference, and set the duty ratio of the switch accordingly. Regulation techniques are discussed in Chap. 7.

The average inductor current must be the same as the average current in the load resistor, since the average capacitor current must be zero for steady-state operation:

$$I_L = I_R = \frac{V_o}{R} \quad (6-10)$$

Since the change in inductor current is known from Eqs. (6-7) and (6-8), the maximum and minimum values of the inductor current are computed as

$$\begin{aligned} I_{\max} &= I_L + \frac{\Delta i_L}{2} \\ &= \frac{V_o}{R} + \frac{1}{2} \left[\frac{V_o}{L} (1 - D)T \right] = V_o \left(\frac{1}{R} + \frac{1 - D}{2Lf} \right) \end{aligned} \quad (6-11)$$

$$\begin{aligned} I_{\min} &= I_L - \frac{\Delta i_L}{2} \\ &= \frac{V_o}{R} - \frac{1}{2} \left[\frac{V_o}{L} (1 - D)T \right] = V_o \left(\frac{1}{R} - \frac{1 - D}{2Lf} \right) \end{aligned} \quad (6-12)$$

where $f = 1/T$ is the switching frequency.

For the preceding analysis to be valid, continuous current in the inductor must be verified. An easy check for continuous current is to calculate the minimum inductor current from Eq. (6-12). Since the minimum value of inductor current must be positive for continuous current, a negative minimum calculated from Eq. (6-12) is not allowed due to the diode and indicates discontinuous current. The circuit will operate for discontinuous inductor current, but the preceding analysis is not valid. Discontinuous-current operation is discussed later in this chapter.

Equation (6-12) can be used to determine the combination of L and f that will result in continuous current. Since $I_{\min} = 0$ is the boundary between continuous and discontinuous current,

$$\begin{aligned} I_{\min} = 0 &= V_o \left(\frac{1}{R} - \frac{1-D}{2Lf} \right) \\ (Lf)_{\min} &= \frac{(1-D)R}{2} \end{aligned} \quad (6-13)$$

If the desired switching frequency is established,

$$L_{\min} = \frac{(1-D)R}{2f} \quad \text{for continuous current} \quad (6-14)$$

where $L_{\min}$ is the minimum inductance required for continuous current. In practice, a value of inductance greater than $L_{\min}$ is desirable to ensure continuous current.

In the design of a buck converter, the peak-to-peak variation in the inductor current is often used as a design criterion. Equation (6-7) can be combined with Eq. (6-9) to determine the value of inductance for a specified peak-to-peak inductor current for continuous-current operation:

$$\Delta i_L = \left(\frac{V_s - V_o}{L} \right) DT = \left(\frac{V_s - V_o}{Lf} \right) D = \frac{V_o(1-D)}{Lf} \quad (6-15)$$

or

$$L = \left(\frac{V_s - V_o}{\Delta i_L f} \right) D = \frac{V_o(1-D)}{\Delta i_L f} \quad (6-16)$$

Since the converter components are assumed to be ideal, the power supplied by the source must be the same as the power absorbed by the load resistor.

$$\begin{aligned} P_s &= P_o \\ V_s I_s &= V_o I_o \end{aligned} \quad (6-17)$$

or

$$\frac{V_o}{V_s} = \frac{I_s}{I_o}$$

Note that the preceding relationship is similar to the voltage-current relationship for a transformer in ac applications. Therefore, the buck converter circuit is equivalent to a dc transformer.

Output Voltage Ripple

In the preceding analysis, the capacitor was assumed to be very large to keep the output voltage constant. In practice, the output voltage cannot be kept perfectly constant with a finite capacitance. The variation in output voltage, or ripple, is computed from the voltage-current relationship of the capacitor. The current in the capacitor is

$$i_C = i_L - i_R$$

shown in Fig. 6-5a.

While the capacitor current is positive, the capacitor is charging. From the definition of capacitance,

$$Q = CV_o$$

$$\Delta Q = C \Delta V_o$$

$$\Delta V_o = \frac{\Delta Q}{C}$$

The change in charge ΔQ is the area of the triangle above the time axis

$$\Delta Q = \frac{1}{2} \left(\frac{T}{2} \right) \left(\frac{\Delta i_L}{2} \right) = \frac{T \Delta i_L}{8}$$

resulting in

$$\Delta V_o = \frac{T \Delta i_L}{8C}$$

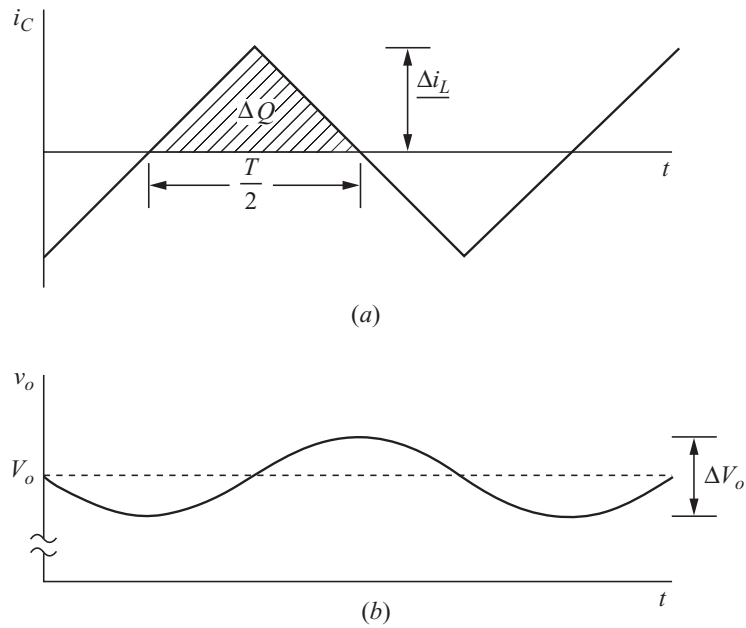


Figure 6-5 Buck converter waveforms. (a) Capacitor current; (b) Capacitor ripple voltage.

Using Eq. (6-8) for Δi_L ,

$$\Delta V_o = \frac{T V_o}{8CL} (1 - D) T = \frac{V_o (1 - D)}{8LCf^2} \quad (6-18)$$

In this equation, ΔV_o is the peak-to-peak ripple voltage at the output, as shown in Fig. 6-5b. It is also useful to express the ripple as a fraction of the output voltage,

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{1 - D}{8LCf^2}} \quad (6-19)$$

In design, it is useful to rearrange the preceding equation to express required capacitance in terms of specified voltage ripple:

$$\boxed{C = \frac{1 - D}{8L(\Delta V_o/V_o)f^2}} \quad (6-20)$$

If the ripple is not large, the assumption of a constant output voltage is reasonable and the preceding analysis is essentially valid.

EXAMPLE 6-1

Buck Converter

The buck dc-dc converter of Fig. 6-3a has the following parameters:

$$\begin{aligned} V_s &= 50 \text{ V} \\ D &= 0.4 \\ L &= 400 \text{ } \mu\text{H} \\ C &= 100 \text{ } \mu\text{F} \\ f &= 20 \text{ kHz} \\ R &= 20 \text{ } \Omega \end{aligned}$$

Assuming ideal components, calculate (a) the output voltage V_o , (b) the maximum and minimum inductor current, and (c) the output voltage ripple.

■ Solution

- (a) The inductor current is assumed to be continuous, and the output voltage is computed from Eq. (6-9),

$$V_o = V_s D = (50)(0.4) = 20 \text{ V}$$

- (b) Maximum and minimum inductor currents are computed from Eqs. (6-11) and (6-12).

$$\begin{aligned} I_{\max} &= V_o \left(\frac{1}{R} + \frac{1 - D}{2Lf} \right) \\ &= 20 \left[\frac{1}{20} + \frac{1 - 0.4}{2(400)(10)^{-6}(20)(10)^3} \right] \\ &= 1 + \frac{1.5}{2} = 1.75 \text{ A} \end{aligned}$$

$$\begin{aligned}
 I_{\min} &= V_o \left(\frac{1}{R} - \frac{1-D}{2Lf} \right) \\
 &= 1 - \frac{1.5}{2} = 0.25 \text{ A}
 \end{aligned}$$

The average inductor current is 1 A, and $\Delta i_L = 1.5$ A. Note that the minimum inductor current is positive, verifying that the assumption of continuous current was valid.

(c) The output voltage ripple is computed from Eq. (6-19).

$$\begin{aligned}
 \frac{\Delta V_o}{V_o} &= \frac{1-D}{8LCf^2} = \frac{1-0.4}{8(400)(10)^{-6}(100)(10)^{-6}(20,000)^2} \\
 &= 0.00469 = 0.469\%
 \end{aligned}$$

Since the output ripple is sufficiently small, the assumption of a constant output voltage was reasonable.

Capacitor Resistance—The Effect on Ripple Voltage

The output voltage ripple in Eqs. (6-18) and (6-19) is based on an ideal capacitor. A real capacitor can be modeled as a capacitance with an equivalent series resistance (ESR) and an equivalent series inductance (ESL). The ESR may have a significant effect on the output voltage ripple, often producing a ripple voltage greater than that of the ideal capacitance. The inductance in the capacitor is usually not a significant factor at typical switching frequencies. Figure 6.6 shows a capacitor model that is appropriate for most applications.

The ripple due to the ESR can be approximated by first determining the current in the capacitor, assuming the capacitor to be ideal. For the buck converter in the continuous-current mode, capacitor current is the triangular current waveform of Fig. 6-4c. The voltage variation across the capacitor resistance is

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = \Delta i_L r_C \quad (6-21)$$

To estimate a worst-case condition, one could assume that the peak-to-peak ripple voltage due to the ESR algebraically adds to the ripple due to the capacitance. However, the peaks of the capacitor and the ESR ripple voltages will not coincide, so

$$\Delta V_o < \Delta V_{o,C} + \Delta V_{o,\text{ESR}} \quad (6-22)$$

where $\Delta V_{o,C}$ is ΔV_o in Eq. (6-18). The ripple voltage due to the ESR can be much larger than the ripple due to the pure capacitance. In that case, the output capacitor is chosen on the basis of the equivalent series resistance rather than capacitance only.

$$\Delta V_o \approx \Delta V_{o,\text{ESR}} = \Delta i_C r_C \quad (6-23)$$

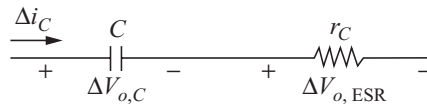


Figure 6-6 A model for the capacitor including the equivalent series resistance (ESR).

Capacitor ESR is inversely proportional to the capacitance value—a larger capacitance results in a lower ESR. Manufacturers provide what are known as *low-ESR capacitors* for power supply applications.

In Example 6-1, the 100- μF capacitor may have an ESR of $r_C = 0.1\ \Omega$. The ripple voltage due to the ESR is calculated as

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = \Delta i_L r_C = (1.5\text{ A})(0.1\ \Omega) = 0.15\text{ V}$$

Expressed as a percent, $\Delta V_o/V_o$ is $0.15/20 = 0.75$ percent. The total ripple can then be approximated as 0.75 percent.

Synchronous Rectification for the Buck Converter

Many buck converters use a second MOSFET in place of the diode. When S_2 is on and S_1 is off, current flows upward out of the drain of S_2 . The advantage of this configuration is that the second MOSFET will have a much lower voltage drop across it compared to a diode, resulting in higher circuit efficiency. This is especially important in low-voltage, high-current applications. A Schottky diode would have a voltage of 0.3 to 0.4 V across it while conducting, whereas a MOSFET will have an extremely low voltage drop due to an R_{DSon} as low as single-digit milliohms. This circuit has a control scheme known as *synchronous switching*, or *synchronous rectification*. The second MOSFET is known as a *synchronous rectifier*. The two MOSFETs must not be on at the same time to prevent a short circuit across the source, so a “dead time” is built into the switching control—one MOSFET is turned off before the other is turned on. A diode is placed in parallel with the second MOSFET to provide a conducting path for inductor current during the dead time when both MOSFETs are off. This diode may be the MOSFET body diode, or it may be an extra diode, most likely a Schottky diode, for improved switching. The synchronous buck converter should be operated in the continuous-current mode because the MOSFET would allow the inductor current to go negative.

Other converter topologies presented in this chapter and in Chap. 7 can utilize MOSFETs in place of diodes.

6.4 DESIGN CONSIDERATIONS

Most buck converters are designed for continuous-current operation. The choice of switching frequency and inductance to give continuous current is given by Eq. (6-13), and the output voltage ripple is described by Eqs. (6-16) and (6-21). Note that as the switching frequency increases, the minimum size of the inductor to produce continuous current and the minimum size of the capacitor to limit output ripple both decrease. Therefore, high switching frequencies are desirable to reduce the size of both the inductor and the capacitor.

The tradeoff for high switching frequencies is increased power loss in the switches, which is discussed later in this chapter and in Chap. 10. Increased power loss in the switches means that heat is produced. This decreases the converter’s efficiency and may require a large heat sink, offsetting the reduction in size of the inductor and capacitor. Typical switching frequencies are above 20 kHz to avoid audio noise, and they extend well into the 100s of kilohertz and into the megahertz range. Some designers consider about 500 kHz to be the best compromise

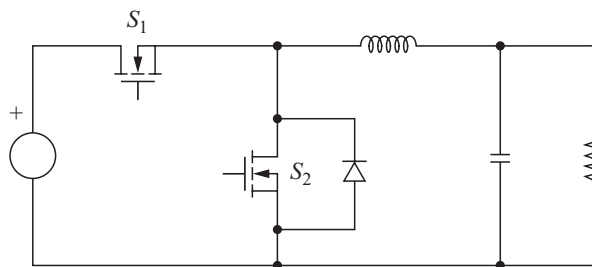


Figure 6-7 A synchronous buck converter. The MOSFET S_2 carries the inductor current when S_1 is off to provide a lower voltage drop than a diode.

between small component size and efficiency. Other designers prefer to use lower switching frequencies of about 50 kHz to keep switching losses small, while still others prefer frequencies larger than 1 MHz. As switching devices improve, switching frequencies will increase.

For low-voltage, high-current applications, the synchronous rectification scheme of Fig. 6-7 is preferred over using a diode for the second switch. The voltage across the conducting MOSFET will be much less than that across a diode, resulting in lower losses.

The inductor value should be larger than $L_{\min}$ in Eq. (6-14) to ensure continuous-current operation. Some designers select a value 25 percent larger than $L_{\min}$. Other designers use different criteria, such as setting the inductor current variation, Δi_L in Eq. (6-15), to a desired value, such as 40 percent of the average inductor current. A smaller Δi_L results in lower peak and rms inductor currents and a lower rms capacitor current but requires a larger inductor.

The inductor wire must be rated at the rms current, and the core should not saturate for peak inductor current. The capacitor must be selected to limit the output ripple to the design specifications, to withstand peak output voltage, and to carry the required rms current.

The switch (usually a MOSFET with a low $R_{DS(on)}$) and diode (or second MOSFET for synchronous rectification) must withstand maximum voltage stress when off and maximum current when on. The temperature ratings must not be exceeded, often requiring a heat sink.

Assuming ideal switches and an ideal inductor in the initial design is usually reasonable. However, the ESR of the capacitor should be included because it typically gives a more significant output voltage ripple than the ideal device and greatly influences the choice of capacitor size.

EXAMPLE 6-2

Buck Converter Design 1

Design a buck converter to produce an output voltage of 18 V across a 10- Ω load resistor. The output voltage ripple must not exceed 0.5 percent. The dc supply is 48 V. Design for continuous inductor current. Specify the duty ratio, the switching frequency, the values of the inductor and capacitor, the peak voltage rating of each device, and the rms current in the inductor and capacitor. Assume ideal components.

■ Solution

Using the buck converter circuit in Fig. 6-3a, the duty ratio for continuous-current operation is determined from Eq. (6-9):

$$D = \frac{V_o}{V_s} = \frac{18}{48} = 0.375$$

The switching frequency and inductor size must be selected for continuous-current operation. Let the switching frequency arbitrarily be 40 kHz, which is well above the audio range and is low enough to keep switching losses small. The minimum inductor size is determined from Eq. (6-14).

$$L_{\min} = \frac{(1-D)(R)}{2f} = \frac{(1-0.375)(10)}{2(40,000)} = 78 \mu\text{H}$$

Let the inductor be 25 percent larger than the minimum to ensure that inductor current is continuous.

$$L = 1.25L_{\min} = (1.25)(78 \mu\text{H}) = 97.5 \mu\text{H}$$

Average inductor current and the change in current are determined from Eqs. (6-10) and (6-17).

$$I_L = \frac{V_o}{R} = \frac{18}{10} = 1.8 \text{ A}$$

$$\Delta i_L = \left(\frac{V_s - V_o}{L} \right) DT = \frac{48 - 18}{97.5(10)^{-6}} (0.375) \left(\frac{1}{40,000} \right) = 2.88 \text{ A}$$

The maximum and minimum inductor currents are determined from Eqs. (6-11) and (6-12).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = 1.8 + 1.44 = 3.24 \text{ A}$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = 1.8 - 1.44 = 0.36 \text{ A}$$

The inductor must be rated for rms current, which is computed as in Chap. 2 (see Example 2-8). For the offset triangular wave,

$$I_{L,\text{rms}} = \sqrt{I_L^2 + \left(\frac{\Delta i_L/2}{\sqrt{3}} \right)^2} = \sqrt{(1.8)^2 + \left(\frac{1.44}{\sqrt{3}} \right)^2} = 1.98 \text{ A}$$

The capacitor is selected using Eq. (6-20).

$$C = \frac{1-D}{8L(\Delta V_o/V_o)f^2} = \frac{1-0.375}{8(97.5)(10)^{-6}(0.005)(40,000)^2} = 100 \mu\text{F}$$

Peak capacitor current is $\Delta i_L/2 = 1.44 \text{ A}$, and rms capacitor current for the triangular waveform is $1.44/\sqrt{3} = 0.83 \text{ A}$. The maximum voltage across the switch and diode is V_s , or 48 V. The inductor voltage when the switch is closed is $V_s - V_o = 48 - 18 = 30 \text{ V}$. The inductor voltage when the switch is open is $V_o = 18 \text{ V}$. Therefore, the inductor must withstand 30 V. The capacitor must be rated for the 18-V output.

EXAMPLE 6-3**Buck Converter Design 2**

Power supplies for telecommunications applications may require high currents at low voltages. Design a buck converter that has an input voltage of 3.3 V and an output voltage of 1.2 V. The output current varies between 4 and 6 A. The output voltage ripple must not exceed 2 percent. Specify the inductor value such that the peak-to-peak variation in inductor current does not exceed 40 percent of the average value. Determine the required rms current rating of the inductor and of the capacitor. Determine the maximum equivalent series resistance of the capacitor.

■ Solution

Because of the low voltage and high output current in this application, the synchronous rectification buck converter of Fig. 6-7 is used. The duty ratio is determined from Eq. (6-9).

$$D = \frac{V_o}{V_s} = \frac{1.2}{3.3} = 0.364$$

The switching frequency and inductor size must be selected for continuous-current operation. Let the switching frequency arbitrarily be 500 kHz to give a good tradeoff between small component size and low switching losses.

The average inductor current is the same as the output current. Analyzing the circuit for an output current of 4 A,

$$I_L = I_o = 4 \text{ A}$$

$$\Delta i_L = (40\%)(4) = 1.6 \text{ A}$$

Using Eq. (6-16),

$$L = \left(\frac{V_s - V_o}{\Delta i_L f} \right) D = \frac{3.3 - 1.2}{(1.6)(500,000)} (0.364) = 0.955 \text{ } \mu\text{H}$$

Analyzing the circuit for an output current of 6 A,

$$I_L = I_o = 6 \text{ A}$$

$$\Delta i_L = (40\%)(6) = 2.4 \text{ A}$$

resulting in

$$L = \left(\frac{V_s - V_o}{\Delta i_L f} \right) D = \frac{3.3 - 1.2}{(2.4)(500,000)} (0.364) = 0.636 \text{ } \mu\text{H}$$

Since 0.636 μH would be too small for the 4-A output, use $L = 0.955 \text{ } \mu\text{H}$, which would be rounded to 1 μH .

Inductor rms current is determined from

$$I_{L,\text{rms}} = \sqrt{I_L^2 + \left(\frac{\Delta i_L/2}{\sqrt{3}} \right)^2}$$

(see Chap. 2). From Eq. (6-15), the variation in inductor current is 1.6 A for each output current. Using the 6-A output current, the inductor must be rated for an rms current of

$$I_{L, \text{rms}} = \sqrt{6^2 + \left(\frac{0.8}{\sqrt{3}}\right)^2} = 6.02 \text{ A}$$

Note that the average inductor current would be a good approximation to the rms current since the variation is relatively small.

Using $L = 1 \mu\text{H}$ in Eq. (6-20), the minimum capacitance is determined as

$$C = \frac{1 - D}{8L(\Delta V_o / V_o)f^2} = \frac{1 - 0.364}{8(1)(10)^{-6}(0.02)(500,000)^2} = 0.16 \mu\text{F}$$

The allowable output voltage ripple of 2 percent is $(0.02)(1.2) = 24 \text{ mV}$. The maximum ESR is computed from Eq. (6-23).

$$\Delta V_o \approx r_C \Delta i_C = r_C \Delta i_L$$

or
$$r_C = \frac{\Delta V_o}{\Delta i_C} = \frac{24 \text{ mV}}{1.6 \text{ A}} = 15 \text{ m}\Omega$$

At this point, the designer would search manufacturer's specifications for a capacitor having 15-m Ω ESR. The capacitor may have to be much larger than the calculated value of 0.16 μF to meet the ESR requirement. Peak capacitor current is $\Delta i_L/2 = 0.8 \text{ A}$, and rms capacitor current for the triangular waveform is $0.8/\sqrt{3} = 0.46 \text{ A}$.

6.5 THE BOOST CONVERTER

The boost converter is shown in Fig. 6-8. This is another switching converter that operates by periodically opening and closing an electronic switch. It is called a boost converter because the output voltage is larger than the input.

Voltage and Current Relationships

The analysis assumes the following:

1. Steady-state conditions exist.
2. The switching period is T , and the switch is closed for time DT and open for $(1-D)T$.
3. The inductor current is continuous (always positive).
4. The capacitor is very large, and the output voltage is held constant at voltage V_o .
5. The components are ideal.

The analysis proceeds by examining the inductor voltage and current for the switch closed and again for the switch open.

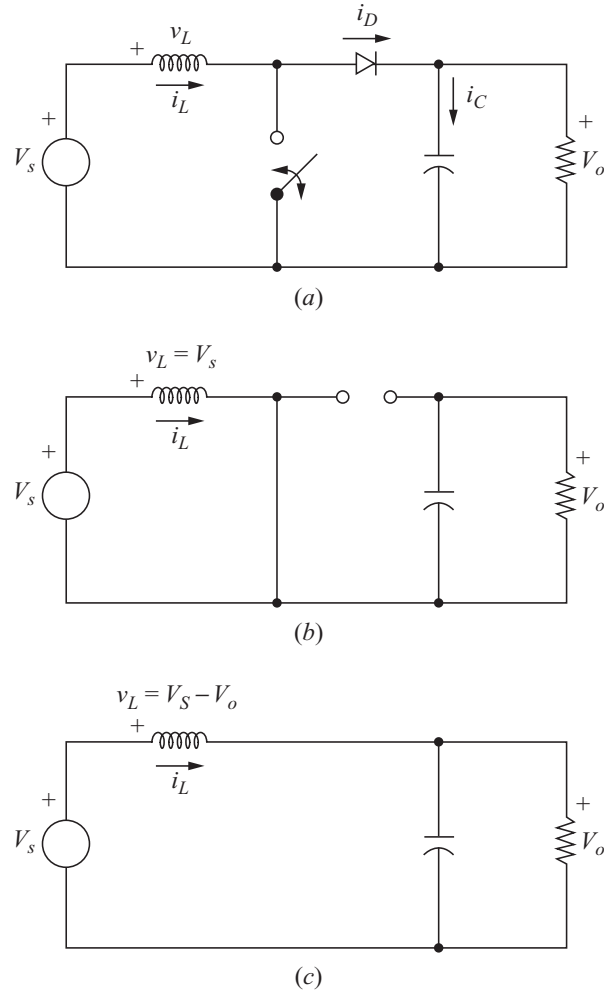


Figure 6-8 The boost converter. (a) Circuit; (b) Equivalent circuit for the switch closed; (c) Equivalent circuit for the switch open.

Analysis for the Switch Closed When the switch is closed, the diode is reverse-biased. Kirchhoff's voltage law around the path containing the source, inductor, and closed switch is

$$v_L = V_s = L \frac{di_L}{dt} \quad \text{or} \quad \frac{di_L}{dt} = \frac{V_s}{L} \quad (6-24)$$

The rate of change of current is a constant, so the current increases linearly while the switch is closed, as shown in Fig. 6-9b. The change in inductor current is computed from

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s}{L}$$

Solving for Δi_L for the switch closed,

$$(\Delta i_L)_{\text{closed}} = \frac{V_s DT}{L} \quad (6-25)$$

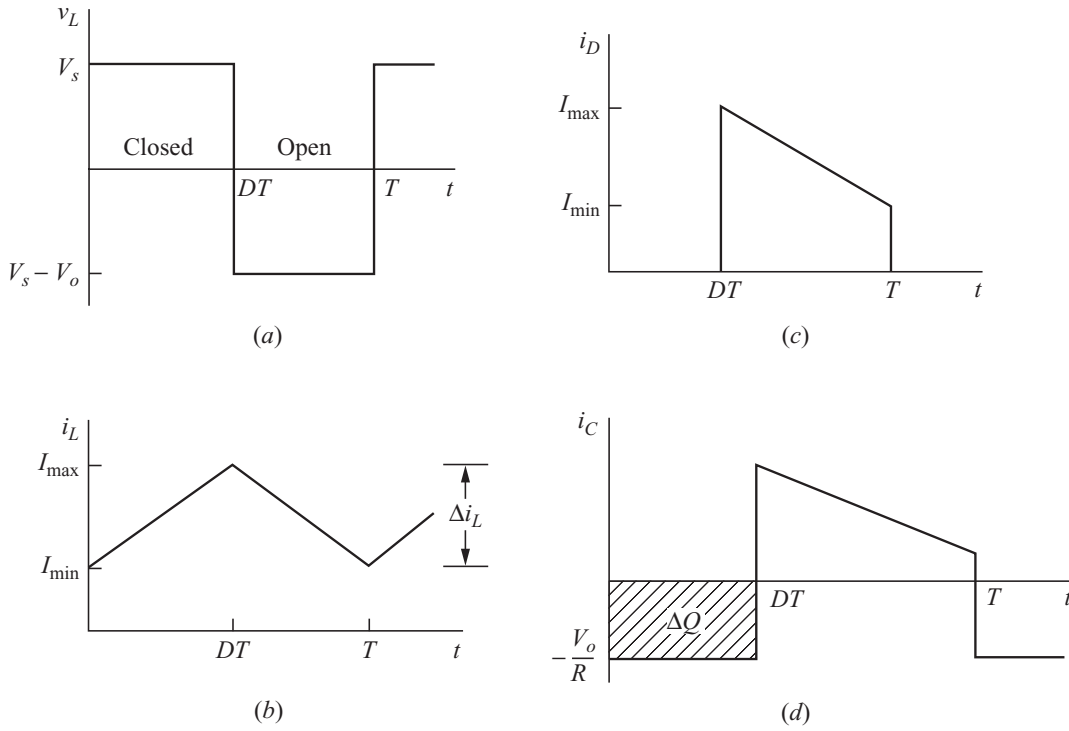


Figure 6-9 Boost converter waveforms. (a) Inductor voltage; (b) Inductor current; (c) Diode current; (d) Capacitor current.

Analysis for the Switch Open When the switch is opened, the inductor current cannot change instantaneously, so the diode becomes forward-biased to provide a path for inductor current. Assuming that the output voltage V_o is a constant, the voltage across the inductor is

$$v_L = V_s - V_o = L \frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_s - V_o}{L}$$

The rate of change of inductor current is a constant, so the current must change linearly while the switch is open. The change in inductor current while the switch is open is

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = \frac{V_s - V_o}{L}$$

Solving for Δi_L ,

$$(\Delta i_L)_{\text{open}} = \frac{(V_s - V_o)(1-D)T}{L} \quad (6-26)$$

For steady-state operation, the net change in inductor current must be zero. Using Eqs. (6-25) and (6-26),

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

$$\frac{V_s D T}{L} + \frac{(V_s - V_o)(1 - D)T}{L} = 0$$

Solving for V_o ,

$$V_s(D + 1 - D) - V_o(1 - D) = 0$$

$$\boxed{V_o = \frac{V_s}{1 - D}} \quad (6-27)$$

Also, the average inductor voltage must be zero for periodic operation. Expressing the average inductor voltage over one switching period,

$$V_L = V_s D + (V_s - V_o)(1 - D) = 0$$

Solving for V_o yields the same result as in Eq. (6-27).

Equation (6-27) shows that if the switch is always open and D is zero, the output voltage is the same as the input. As the duty ratio is increased, the denominator of Eq. (6-27) becomes smaller, resulting in a larger output voltage. *The boost converter produces an output voltage that is greater than or equal to the input voltage.* However, the output voltage cannot be less than the input, as was the case with the buck converter.

As the duty ratio of the switch approaches 1, the output voltage goes to infinity according to Eq. (6-27). However, Eq. (6-27) is based on ideal components. Real components that have losses will prevent such an occurrence, as shown later in this section. Figure 6-9 shows the voltage and current waveforms for the boost converter.

The average current in the inductor is determined by recognizing that the average power supplied by the source must be the same as the average power absorbed by the load resistor. Output power is

$$P_o = \frac{V_o^2}{R} = V_o I_o$$

and input power is $V_s I_s = V_s I_L$. Equating input and output powers and using Eq. (6-27),

$$V_s I_L = \frac{V_o^2}{R} = \frac{[V_s/(1 - D)]^2}{R} = \frac{V_s^2}{(1 - D)^2} R$$

By solving for average inductor current and making various substitutions, I_L can be expressed as

$$I_L = \frac{V_s}{(1-D)^2 R} = \frac{V_o^2}{V_s R} = \frac{V_o I_o}{V_s} \quad (6-28)$$

Maximum and minimum inductor currents are determined by using the average value and the change in current from Eq. (6-25).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} + \frac{V_s D T}{2L} \quad (6-29)$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} - \frac{V_s D T}{2L} \quad (6-30)$$

Equation (6-27) was developed with the assumption that the inductor current is continuous, meaning that it is always positive. A condition necessary for continuous inductor current is for $I_{\min}$ to be positive. Therefore, the boundary between continuous and discontinuous inductor current is determined from

$$I_{\min} = 0 = \frac{V_s}{(1-D)^2 R} - \frac{V_s D T}{2L}$$

or

$$\frac{V_s}{(1-D)^2 R} = \frac{V_s D T}{2L} = \frac{V_s D}{2Lf}$$

The minimum combination of inductance and switching frequency for continuous current in the boost converter is therefore

$$(Lf)_{\min} = \frac{D(1-D)^2 R}{2} \quad (6-31)$$

or

$$L_{\min} = \frac{D(1-D)^2 R}{2f} \quad (6-32)$$

A boost converter designed for continuous-current operation will have an inductor value greater than $L_{\min}$.

From a design perspective, it is useful to express L in terms of a desired Δi_L ,

$$L = \frac{V_s D T}{\Delta i_L} = \frac{V_s D}{\Delta i_L f} \quad (6-33)$$

Output Voltage Ripple

The preceding equations were developed on the assumption that the output voltage was a constant, implying an infinite capacitance. In practice, a finite capacitance will result in some fluctuation in output voltage, or ripple.

The peak-to-peak output voltage ripple can be calculated from the capacitor current waveform, shown in Fig. 6-9d. The change in capacitor charge can be calculated from

$$|\Delta Q| = \left(\frac{V_o}{R}\right)DT = C\Delta V_o$$

An expression for ripple voltage is then

$$\Delta V_o = \frac{V_o DT}{RC} = \frac{V_o D}{RCf}$$

or

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{D}{RCf}} \quad (6-34)$$

where f is the switching frequency. Alternatively, expressing capacitance in terms of output voltage ripple yields

$$C = \frac{D}{R(\Delta V_o/V_o)f} \quad (6-35)$$

As with the buck converter, equivalent series resistance of the capacitor can contribute significantly to the output voltage ripple. The peak-to-peak variation in capacitor current (Fig. 6-9) is the same as the maximum current in the inductor. The voltage ripple due to the ESR is

$$\Delta V_{o, \text{ESR}} = \Delta i_C r_C = I_{L, \text{max}} r_C \quad (6-36)$$

EXAMPLE 6-4

Boost Converter Design 1

Design a boost converter that will have an output of 30 V from a 12-V source. Design for continuous inductor current and an output ripple voltage of less than one percent. The load is a resistance of 50 Ω . Assume ideal components for this design.

■ Solution

First, determine the duty ratio from Eq. (6-27),

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{12}{30} = 0.6$$

If the switching frequency is selected at 25 kHz to be above the audio range, then the minimum inductance for continuous current is determined from Eq. (6-32).

$$L_{\min} = \frac{D(1-D)^2(R)}{2f} = \frac{0.6(1-0.6)^2(50)}{2(25,000)} = 96 \mu\text{H}$$

To provide a margin to ensure continuous current, let $L = 120 \mu\text{H}$. Note that L and f are selected somewhat arbitrarily and that other combinations will also give continuous current.

Using Eqs. (6-28) and (6-25),

$$I_L = \frac{V_s}{(1-D)^2(R)} = \frac{12}{(1-0.6)^2(50)} = 1.5 \text{ A}$$

$$\frac{\Delta i_L}{2} = \frac{V_s D T}{2L} = \frac{(12)(0.6)}{(2)(120)(10)^{-6}(25,000)} = 1.2 \text{ A}$$

$$I_{\max} = 1.5 + 1.2 = 2.7 \text{ A}$$

$$I_{\min} = 1.5 - 1.2 = 0.3 \text{ A}$$

The minimum capacitance required to limit the output ripple voltage to 1 percent is determined from Eq. (6-35).

$$C \geq \frac{D}{R(\Delta V_o/V_o)f} = \frac{0.6}{(50)(0.01)(25,000)} = 48 \text{ } \mu\text{F}$$

EXAMPLE 6-5

Boost Converter Design 2

A boost converter is required to have an output voltage of 8 V and supply a load current of 1 A. The input voltage varies from 2.7 to 4.2 V. A control circuit adjusts the duty ratio to keep the output voltage constant. Select the switching frequency. Determine a value for the inductor such that the variation in inductor current is no more than 40 percent of the average inductor current for all operating conditions. Determine a value of an ideal capacitor such that the output voltage ripple is no more than 2 percent. Determine the maximum capacitor equivalent series resistance for a 2 percent ripple.

■ Solution

Somewhat arbitrarily, choose 200 kHz for the switching frequency. The circuit must be analyzed for both input voltage extremes to determine the worst-case condition. For $V_s = 2.7 \text{ V}$, the duty ratio is determined from Eq. (6-27).

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{2.7}{8} = 0.663$$

Average inductor current is determined from Eq. (6-28).

$$I_L = \frac{V_o I_o}{V_s} = \frac{8(1)}{2.7} = 2.96 \text{ A}$$

The variation in inductor current to meet the 40 percent specification is then $\Delta i_L = 0.4(2.96) = 1.19 \text{ A}$. The inductance is then determined from Eq. (6-33).

$$L = \frac{V_s D}{\Delta i_L f} = \frac{2.7(0.663)}{1.19(200,000)} = 7.5 \text{ } \mu\text{H}$$

Repeating the calculations for $V_s = 4.2 \text{ V}$,

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{4.2}{8} = 0.475$$

$$I_L = \frac{V_o I_o}{V_s} = \frac{8(1)}{4.2} = 1.90 \text{ A}$$

The variation in inductor current for this case is $\Delta i_L = 0.4(1.90) = 0.762 \text{ A}$, and

$$L = \frac{V_s D}{\Delta i_L f} = \frac{4.2(0.475)}{0.762(200,000)} = 13.1 \mu\text{H}$$

The inductor must be 13.1 μH to satisfy the specifications for the total range of input voltages.

Equation (6-35), using the maximum value of D , gives the minimum capacitance as

$$C = \frac{D}{R(\Delta V_o/V_o)f} = \frac{D}{(V_o/I_o)(\Delta V_o/V_o)f} = \frac{0.663}{(8/1)(0.02)(200,000)} = 20.7 \mu\text{F}$$

The maximum ESR is determined from Eq. (6-36), using the maximum peak-to-peak variation in capacitor current. The peak-to-peak variation in capacitor current is the same as maximum inductor current. The average inductor current varies from 2.96 A at $V_s = 2.7$ V to 1.90 A at $V_s = 4.2$ V. The variation in inductor current is 0.762 A for $V_s = 4.2$ A, but it must be recalculated for $V_s = 2.7$ V using the 13.1- μH value selected, yielding

$$\Delta i_L = \frac{V_s D}{L f} = \frac{2.7(0.663)}{13.1(10)^{-6}(200,000)} = 0.683 \text{ A}$$

Maximum inductor current for each case is then computed as

$$I_{L, \max, 2.7\text{V}} = I_L + \frac{\Delta i_L}{2} = 2.96 + \frac{0.683}{2} = 3.30 \text{ A}$$

$$I_{L, \max, 4.2\text{V}} = I_L + \frac{\Delta i_L}{2} = 1.90 + \frac{0.762}{2} = 2.28 \text{ A}$$

This shows that the largest peak-to-peak current variation in the capacitor will be 3.30 A. The output voltage ripple due to the capacitor ESR must be no more than $(0.02)(8) = 0.16$ V. Using Eq. (6-36),

$$\Delta V_{o, \text{ESR}} = \Delta i_C r_C = I_{L, \max} r_C = 3.3 r_C = 0.16 \text{ V}$$

which gives

$$r_C = \frac{0.16 \text{ V}}{3.3 \text{ A}} = 48 \text{ m}\Omega$$

In practice, a capacitor that has an ESR of 48 $\text{m}\Omega$ or less could have a capacitance value much larger than the 20.7 μF calculated.

Inductor Resistance

Inductors should be designed to have small resistance to minimize power loss and maximize efficiency. The existence of a small inductor resistance does not substantially change the analysis of the buck converter as presented previously in this chapter. However, inductor resistance affects performance of the boost converter, especially at high duty ratios.

For the boost converter, recall that the output voltage for the ideal case is

$$V_o = \frac{V_s}{1 - D} \quad (6-37)$$

To investigate the effect of inductor resistance on the output voltage, assume that the inductor current is approximately constant. The source current is the same as the inductor current, and average diode current is the same as average load current. The power supplied by the source must be the same as the power absorbed by the load and the inductor resistance, neglecting other losses.

$$\begin{aligned} P_s &= P_o + P_{r_L} \\ V_s I_L &= V_o I_D + I_L^2 r_L \end{aligned} \quad (6-38)$$

where r_L is the series resistance of the inductor. The diode current is equal to the inductor current when the switch is off and is zero when the switch is on. Therefore, the average diode current is

$$I_D = I_L(1 - D) \quad (6-39)$$

Substituting for I_D into Eq. (6-38),

$$V_s I_L = V_o I_L(1 - D) + I_L^2 r_L$$

which becomes

$$V_s = V_o(1 - D) + I_L r_L \quad (6-40)$$

In terms of V_o from Eq. (6-39), I_L is

$$I_L = \frac{I_D}{1 - D} = \frac{V_o/R}{1 - D} \quad (6-41)$$

Substituting for I_L into Eq. (6-40),

$$V_s = \frac{V_o r_L}{R(1 - D)} + V_o(1 - D)$$

Solving for V_o ,

$$V_o = \left(\frac{V_s}{1 - D} \right) \left(\frac{1}{1 + r_L/[R(1 - D)^2]} \right) \quad (6-42)$$

The preceding equation is similar to that for an ideal converter but includes a correction factor to account for the inductor resistance. Figure 6-10a shows the output voltage of the boost converter with and without inductor resistance.

The inductor resistance also has an effect on the power efficiency of converters. Efficiency is the ratio of output power to output power plus losses. For the boost converter

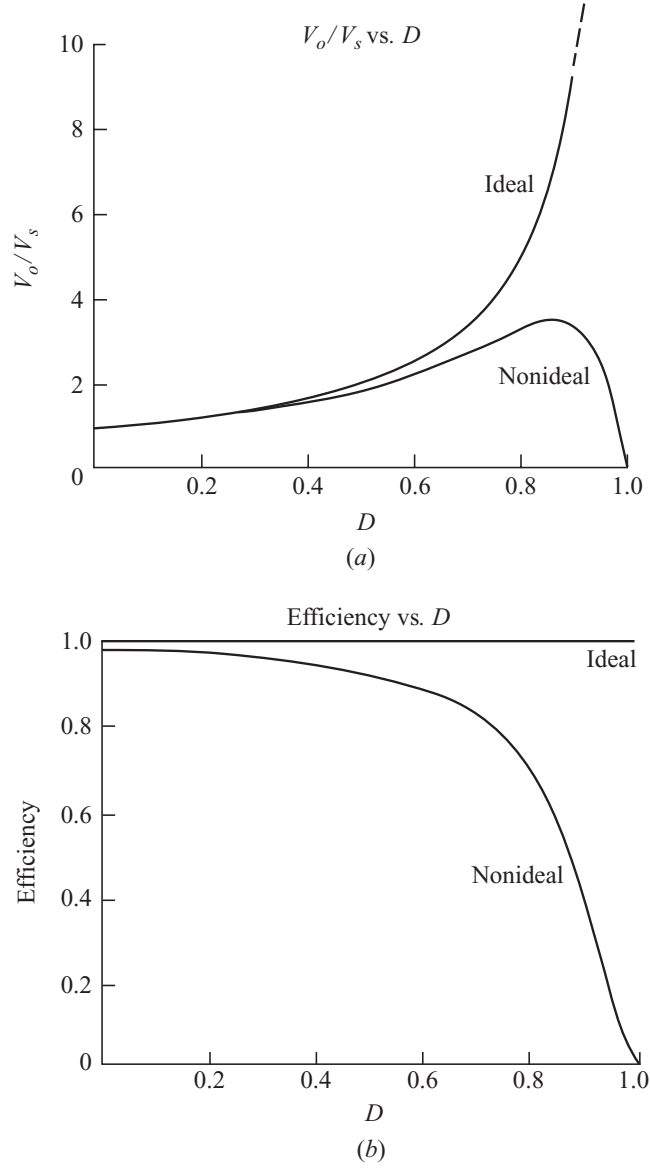


Figure 6-10 Boost converter for a nonideal inductor.
(a) Output voltage; (b) Boost converter efficiency.

$$\eta = \frac{P_o}{P_o + P_{\text{loss}}} = \frac{V_o^2/R}{V_o^2/R + I_L^2 r_L} \quad (6-43)$$

Using Eq. (6-41) for I_L ,

$$\eta = \frac{V_o^2/R}{V_o^2/R + (V_o/R)^2/(1-D)r_L} = \frac{1}{1 + r_L[R(1-D)^2]} \quad (6-44)$$

As the duty ratio increases, the efficiency of the boost converter decreases, as indicated in Fig. 6-10b.

6.6 THE BUCK-BOOST CONVERTER

Another basic switched-mode converter is the buck-boost converter shown in Fig. 6-11. The output voltage of the buck-boost converter can be either higher or lower than the input voltage.

Voltage and Current Relationships

Assumptions made about the operation of the converter are as follows:

1. The circuit is operating in the steady state.
2. The inductor current is continuous.
3. The capacitor is large enough to assume a constant output voltage.
4. The switch is closed for time DT and open for $(1-D)T$.
5. The components are ideal.

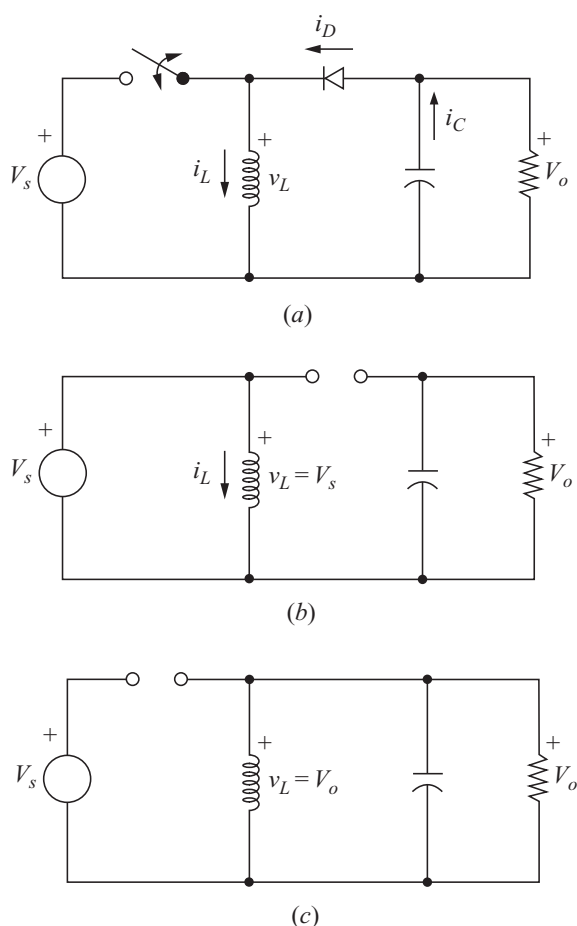


Figure 6-11 Buck-boost converter. (a) Circuit; (b) Equivalent circuit for the switch closed; (c) Equivalent circuit for the switch open.

Analysis for the Switch Closed When the switch is closed, the voltage across the inductor is

$$v_L = V_s = L \frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_s}{L}$$

The rate of change of inductor current is a constant, indicating a linearly increasing inductor current. The preceding equation can be expressed as

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s}{L}$$

Solving for Δi_L when the switch is closed gives

$$(\Delta i_L)_{\text{closed}} = \frac{V_s DT}{L} \quad (6-45)$$

Analysis for the Switch Open When the switch is open, the current in the inductor cannot change instantaneously, resulting in a forward-biased diode and current into the resistor and capacitor. In this condition, the voltage across the inductor is

$$v_L = V_o = L \frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_o}{L}$$

Again, the rate of change of inductor current is constant, and the change in current is

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = \frac{V_o}{L}$$

Solving for Δi_L ,

$$(\Delta i_L)_{\text{open}} = \frac{V_o(1-D)T}{L} \quad (6-46)$$

For steady-state operation, the net change in inductor current must be zero over one period. Using Eqs. (6-45) and (6-46),

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

$$\frac{V_s DT}{L} + \frac{V_o(1-D)T}{L} = 0$$

Solving for V_o ,

$$V_o = -V_s \left(\frac{D}{1-D} \right) \quad (6-47)$$

The required duty ratio for specified input and output voltages can be expressed as

$$D = \frac{|V_o|}{V_s + |V_o|} \quad (6-48)$$

The average inductor voltage is zero for periodic operation, resulting in

$$V_L = V_s D + V_o(1 - D) = 0$$

Solving for V_o yields the same result as Eq. (6-47).

Equation (6-47) shows that the output voltage has opposite polarity from the source voltage. *Output voltage magnitude of the buck-boost converter can be less than that of the source or greater than the source, depending on the duty ratio of the switch.* If $D > 0.5$, the output voltage is larger than the input; and if $D < 0.5$, the output is smaller than the input. Therefore, this circuit combines the capabilities of the buck and boost converters. Polarity reversal on the output may be a disadvantage in some applications, however. Voltage and current waveforms are shown in Fig. 6-12.

Note that the source is never connected directly to the load in the buck-boost converter. Energy is stored in the inductor when the switch is closed and transferred to the load when the switch is open. Hence, the buck-boost converter is also referred to as an *indirect* converter.

Power absorbed by the load must be the same as that supplied by the source, where

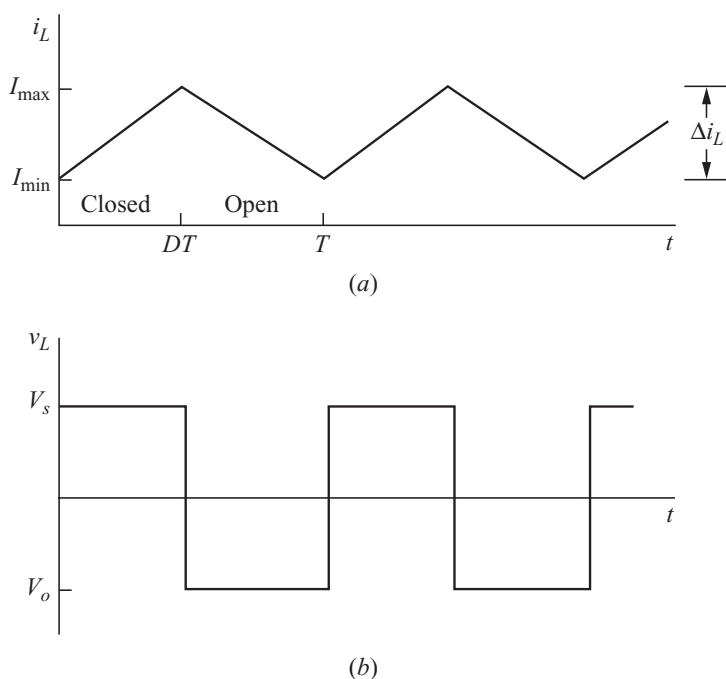


Figure 6-12 Buck-boost converter waveforms.

(a) Inductor current; (b) Inductor voltage; (c) Diode current; (d) Capacitor current.

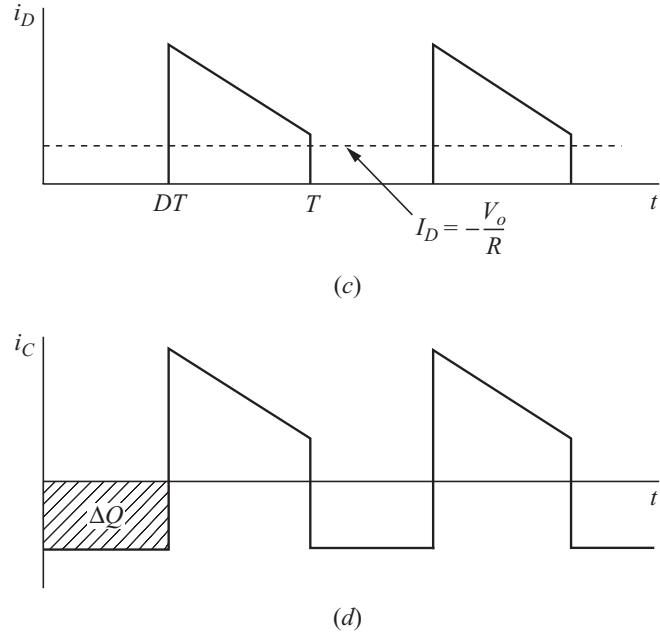


Figure 6-12 (continued)

$$P_o = \frac{V_o^2}{R}$$

$$P_s = V_s I_s$$

$$\frac{V_o^2}{R} = V_s I_s$$

Average source current is related to average inductor current by

$$I_s = I_L D$$

resulting in

$$\frac{V_o^2}{R} = V_s I_L D$$

Substituting for V_o using Eq. (6-47) and solving for I_L , we find

$$I_L = \frac{V_o^2}{V_s R D} = \frac{P_o}{V_s D} = \frac{V_s D}{R(1-D)^2} \quad (6-49)$$

Maximum and minimum inductor currents are determined using Eqs. (6-45) and (6-49).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_s D}{R(1-D)^2} + \frac{V_s D T}{2L} \quad (6-50)$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_s D}{R(1-D)^2} - \frac{V_s D T}{2L} \quad (6-51)$$

For continuous current, the inductor current must remain positive. To determine the boundary between continuous and discontinuous current, $I_{\min}$ is set to zero in Eq. (6-51), resulting in

$$(Lf)_{\min} = \frac{(1-D)^2 R}{2} \quad (6-52)$$

or

$$L_{\min} = \frac{(1-D)^2 R}{2f} \quad (6-53)$$

where f is the switching frequency.

Output Voltage Ripple

The output voltage ripple for the buck-boost converter is computed from the capacitor current waveform of Fig. 6-12d.

$$|\Delta Q| = \left(\frac{V_o}{R} \right) DT = C \Delta V_o$$

Solving for ΔV_o ,

$$\Delta V_o = \frac{V_o DT}{RC} = \frac{V_o D}{RCf}$$

or

$$\frac{\Delta V_o}{V_o} = \frac{D}{RCf} \quad (6-54)$$

As is the case with other converters, the equivalent series resistance of the capacitor can contribute significantly to the output ripple voltage. The peak-to-peak variation in capacitor current is the same as the maximum inductor current. Using the capacitor model shown in Fig. 6-6, where $I_{L,\max}$ is determined from Eq. (6-50),

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = I_{L,\max} r_C \quad (6-55)$$

EXAMPLE 6-6

Buck-Boost Converter

The buck-boost circuit of Fig. 6-11 has these parameters:

$$\begin{aligned} V_s &= 24 \text{ V} \\ D &= 0.4 \\ R &= 5 \, \Omega \\ L &= 20 \, \mu\text{H} \\ C &= 80 \, \mu\text{F} \\ f &= 100 \text{ kHz} \end{aligned}$$

Determine the output voltage, inductor current average, maximum and minimum values, and the output voltage ripple.

■ **Solution**

Output voltage is determined from Eq. (6-47).

$$V_o = -V_s \left(\frac{D}{1-D} \right) = -24 \left(\frac{0.4}{1-0.4} \right) = -16 \text{ V}$$

Inductor current is described by Eqs. (6-49) to (6-51).

$$I_L = \frac{V_s D}{R(1-D)^2} = \frac{24(0.4)}{5(1-0.4)^2} = 5.33 \text{ A}$$

$$\Delta i_L = \frac{V_s D T}{L} = \frac{24(0.4)}{20(10)^{-6}(100,000)} = 4.8 \text{ A}$$

$$I_{L, \max} = I_L + \frac{\Delta i_L}{2} = 5.33 + \frac{4.8}{2} = 7.33 \text{ A}$$

$$I_{L, \min} = I_L - \frac{\Delta i_L}{2} = 5.33 - \frac{4.8}{2} = 2.93 \text{ A}$$

Continuous current is verified by $I_{\min} > 0$.

Output voltage ripple is determined from Eq. (6-54).

$$\frac{\Delta V_o}{V_o} = \frac{D}{RCf} = \frac{0.4}{(5)(80)(10)^{-6}(100,000)} = 0.01 = 1\%$$

6.7 THE ĆUK CONVERTER

The Ćuk switching topology is shown in Fig. 6-13a. Output voltage magnitude can be either larger or smaller than that of the input, and there is a polarity reversal on the output.

The inductor on the input acts as a filter for the dc supply to prevent large harmonic content. Unlike the previous converter topologies where energy transfer is associated with the inductor, energy transfer for the Ćuk converter depends on the capacitor C_1 .

The analysis begins with these assumptions:

1. Both inductors are very large and the currents in them are constant.
2. Both capacitors are very large and the voltages across them are constant.
3. The circuit is operating in steady state, meaning that voltage and current waveforms are periodic.
4. For a duty ratio of D , the switch is closed for time DT and open for $(1-D)T$.
5. The switch and the diode are ideal.

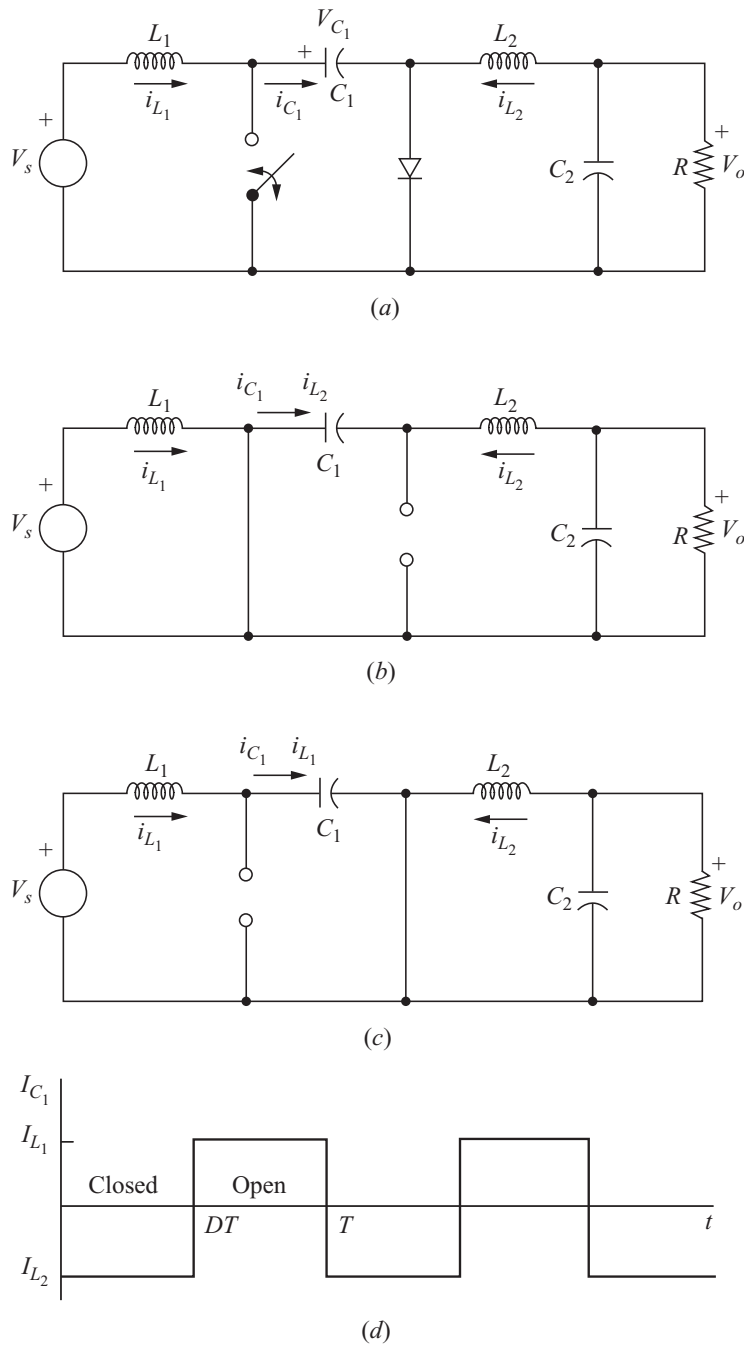


Figure 6-13 The Ćuk converter. (a) Circuit; (b) Equivalent circuit for the switch closed; (c) Equivalent circuit for the switch open; (d) Current in L_1 for a large inductance.

The average voltage across C_1 is computed from Kirchhoff's voltage law around the outermost loop. The average voltage across the inductors is zero for steady-state operation, resulting in

$$V_{C_1} = V_s - V_o$$

With the switch closed, the diode is off and the current in capacitor C_1 is

$$(i_{C_1})_{\text{closed}} = -I_{L_2} \quad (6-56)$$

With the switch open, the currents in L_1 and L_2 force the diode on. The current in capacitor C_1 is

$$(i_{C_1})_{\text{open}} = I_{L_1} \quad (6-57)$$

The power absorbed by the load is equal to the power supplied by the source:

$$-V_o I_{L_2} = V_s I_{L_1} \quad (6-58)$$

For periodic operation, the average capacitor current is zero. With the switch on for time DT and off for $(1-D)T$,

$$[(i_{C_1})_{\text{closed}}]DT + [(i_{C_1})_{\text{open}}](1-D)T = 0$$

Substituting using Eqs. (6-56) and (6-57),

$$-I_{L_2}DT + I_{L_1}(1-D)T = 0 \quad (6-59)$$

or

$$\frac{I_{L_1}}{I_{L_2}} = \frac{D}{1-D}$$

Next, the average power supplied by the source must be the same as the average power absorbed by the load,

$$P_s = P_o$$

$$V_s I_{L_1} = -V_o I_{L_2} \quad (6-60)$$

$$\frac{I_{L_1}}{I_{L_2}} = -\frac{V_o}{V_s}$$

Combining Eqs. (6-59) and (6-60), the relationship between the output and input voltages is

$$V_o = -V_s \left(\frac{D}{1-D} \right) \quad (6-61)$$

The negative sign indicates a polarity reversal between output and input.

Note that the components on the output (L_2 , C_2 , and R) are in the same configuration as the buck converter and that the inductor current has the same form as for the buck converter. Therefore, the ripple, or variation in output voltage, is the same as for the buck converter:

$$\frac{\Delta V_o}{V_o} = \frac{1-D}{8L_2 C_2 f^2} \quad (6-62)$$

The output ripple voltage will be affected by the equivalence series resistance of the capacitor as it was in the convertors discussed previously.

The ripple in C_1 can be estimated by computing the change in v_{C1} in the interval when the switch is open and the currents i_{L1} and i_{C1} are the same. Assuming the current in L_1 to be constant at a level I_{L1} and using Eqs. (6-60) and (6-61), we have

$$\Delta v_{C1} \approx \frac{1}{C_1} \int_{DT}^T I_{L1} d(t) = \frac{I_{L1}}{C_1} (1-D)T = \frac{V_s}{RC_1 f} \left(\frac{D^2}{1-D} \right)$$

or

$$\Delta v_{C1} \approx \frac{V_o D}{RC_1 f} \quad (6-63)$$

The fluctuations in inductor currents can be computed by examining the inductor voltages while the switch is closed. The voltage across L_1 with the switch closed is

$$v_{L1} = V_s = L_1 \frac{di_{L1}}{dt} \quad (6-64)$$

In the time interval DT when the switch is closed, the change in inductor current is

$$\frac{\Delta i_{L1}}{DT} = \frac{V_s}{L_1}$$

or

$$\Delta i_{L1} = \frac{V_s DT}{L_1} = \frac{V_s D}{L_1 f} \quad (6-65)$$

For inductor L_2 , the voltage across it when the switch is closed is

$$v_{L2} = V_o + (V_s - V_o) = V_s = L_2 \frac{di_{L2}}{dt} \quad (6-66)$$

The change in i_{L2} is then

$$\Delta i_{L2} = \frac{V_s DT}{L_2} = \frac{V_s D}{L_2 f} \quad (6-67)$$

For continuous current in the inductors, the average current must be greater than one-half the change in current. Minimum inductor sizes for continuous current are

$$\begin{aligned} L_{1,\min} &= \frac{(1-D)^2 R}{2Df} \\ L_{2,\min} &= \frac{(1-D)R}{2f} \end{aligned} \quad (6-68)$$

EXAMPLE 6-7

Ćuk Converter Design

A Ćuk converter has an input of 12 V and is to have an output of -18 V supplying a 40-W load. Select the duty ratio, the switching frequency, the inductor sizes such that the change in inductor currents is no more than 10 percent of the average inductor current, the output ripple voltage is no more than 1 percent, and the ripple voltage across C_1 is no more than 5 percent.

■ Solution

The duty ratio is obtained from Eq. (6-61),

$$\frac{V_o}{V_s} = -\frac{D}{1-D} = \frac{-18}{12} = -1.5$$

or

$$D = 0.6$$

Next, the switching frequency needs to be selected. Higher switching frequencies result in smaller current variations in the inductors. Let $f = 50$ kHz. The average inductor currents are determined from the power and voltage specifications.

$$I_{L_2} = \frac{P_o}{-V_o} = \frac{40 \text{ W}}{18 \text{ V}} = 2.22 \text{ A}$$

$$I_{L_1} = \frac{P_s}{V_s} = \frac{40 \text{ W}}{12 \text{ V}} = 3.33 \text{ A}$$

The change in inductor currents is computed from Eqs. (6-65) and (6-67).

$$\Delta i_L = \frac{V_s D}{L f}$$

The 10 percent limit in changes in inductor currents requires

$$L_2 \geq \frac{V_s D}{f \Delta i_{L_2}} = \frac{(12)(0.6)}{(50,000)(0.222)} = 649 \text{ } \mu\text{H}$$

$$L_1 \geq \frac{V_s D}{f \Delta i_{L_1}} = \frac{(12)(0.6)}{(50,000)(0.333)} = 432 \text{ } \mu\text{H}$$

From Eq. (6-62), the output ripple specification requires

$$C_2 \geq \frac{1-D}{(\Delta V_o/V_o) 8 L_2 f^2} = \frac{1-0.6}{(0.01)(8)(649)(10)^{-6}(50,000)^2} = 3.08 \text{ } \mu\text{F}$$

Average voltage across C_1 is $V_s - V_o = 12 - (-18) = 30$ V, so the maximum change in v_{C_1} is $(30)(0.05) = 1.5$ V.

The equivalent load resistance is

$$R = \frac{V_o^2}{P} = \frac{(18)^2}{40} = 8.1 \text{ } \Omega$$

Now C_1 is computed from the ripple specification and Eq. (6-63).

$$C_1 \geq \frac{V_o D}{R f \Delta v_{C_1}} = \frac{(18)(0.6)}{(8.1)(50,000)(1.5)} = 17.8 \text{ } \mu\text{F}$$

6.8 THE SINGLE-ENDED PRIMARY INDUCTANCE CONVERTER (SEPIC)

A converter similar to the Ćuk is the single-ended primary inductance converter (SEPIC), as shown in Fig. 6-14. The SEPIC can produce an output voltage that is either greater or less than the input but with no polarity reversal.

To derive the relationship between input and output voltages, these initial assumptions are made:

1. Both inductors are very large and the currents in them are constant.
2. Both capacitors are very large and the voltages across them are constant.
3. The circuit is operating in the steady state, meaning that voltage and current waveforms are periodic.
4. For a duty ratio of D , the switch is closed for time DT and open for $(1 - D)T$.
5. The switch and the diode are ideal.

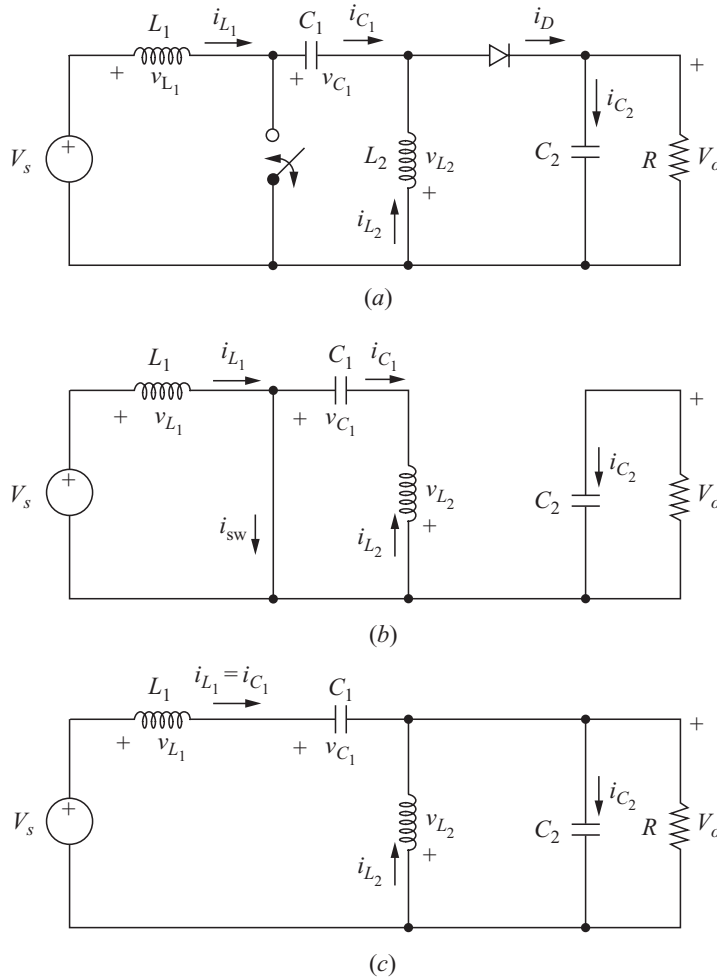


Figure 6-14 (a) SEPIC circuit; (b) Circuit with the switch closed and the diode off; (c) Circuit with the switch open and the diode on.

The inductor current and capacitor voltage restrictions will be removed later to investigate the fluctuations in currents and voltages. The inductor currents are assumed to be continuous in this analysis. Other observations are that the average inductor voltages are zero and that the average capacitor currents are zero for steady-state operation.

Kirchhoff's voltage law around the path containing V_s , L_1 , C_1 , and L_2 gives

$$-V_s + v_{L_1} + v_{C_1} - v_{L_2} = 0$$

Using the average of these voltages,

$$-V_s + 0 + V_{C_1} - 0 = 0$$

showing that the average voltage across the capacitor C_1 is

$$V_{C_1} = V_s \quad (6-69)$$

When the switch is closed, the diode is off, and the circuit is as shown in Fig. 6-14*b*. The voltage across L_1 for the interval DT is

$$v_{L_1} = V_s \quad (6-70)$$

When the switch is open, the diode is on, and the circuit is as shown in Fig. 6-14*c*. Kirchhoff's voltage law around the outermost path gives

$$-V_s + v_{L_1} + v_{C_1} + V_o = 0 \quad (6-71)$$

Assuming that the voltage across C_1 remains constant at its average value of V_s [Eq. (6-69)],

$$-V_s + v_{L_1} + V_s + V_o = 0 \quad (6-72)$$

or

$$v_{L_1} = -V_o \quad (6-73)$$

for the interval $(1 - D)T$. Since the average voltage across an inductor is zero for periodic operation, Eqs. (6-70) and (6-73) are combined to get

$$(v_{L_1, \text{sw closed}})(DT) + (v_{L_1, \text{sw open}})(1 - D)T = 0$$

$$V_s(DT) - V_o(1 - D)T = 0$$

where D is the duty ratio of the switch. The result is

$$\boxed{V_o = V_s \left(\frac{D}{1 - D} \right)} \quad (6-74)$$

which can be expressed as

$$\boxed{D = \frac{V_o}{V_o + V_s}} \quad (6-75)$$

This result is similar to that of the buck-boost and Ćuk converter equations, with the important distinction that there is no polarity reversal between input and output voltages. The ability to have an output voltage greater or less than the input with no polarity reversal makes this converter suitable for many applications.

Assuming no losses in the converter, the power supplied by the source is the same as the power absorbed by the load.

$$P_s = P_o$$

Power supplied by the dc source is voltage times the average current, and the source current is the same as the current in L_1 .

$$P_s = V_s I_s = V_s I_{L1}$$

Output power can be expressed as

$$P_o = V_o I_o$$

resulting in

$$V_s I_{L1} = V_o I_o$$

Solving for average inductor current, which is also the average source current,

$$I_{L1} = I_s = \frac{V_o I_o}{V_s} = \frac{V_o^2}{V_s R} \quad (6-76)$$

The variation in i_{L1} when the switch is closed is found from

$$v_{L1} = V_s = L_1 \left(\frac{di_{L1}}{dt} \right) = L_1 \left(\frac{\Delta i_{L1}}{\Delta t} \right) = L_1 \left(\frac{\Delta i_{L1}}{DT} \right) \quad (6-77)$$

Solving for Δi_{L1} ,

$$\Delta i_{L1} = \frac{V_s DT}{L_1} = \frac{V_s D}{L_1 f} \quad (6-78)$$

For L_2 , the average current is determined from Kirchhoff's current law at the node where C_1 , L_2 , and the diode are connected.

$$i_{L2} = i_D - i_{C1}$$

Diode current is

$$i_D = i_{C2} + I_o$$

which makes

$$i_{L2} = i_{C2} + I_o - i_{C1}$$

The average current in each capacitor is zero, so the average current in L_2 is

$$I_{L2} = I_o \quad (6-79)$$

The variation in i_{L2} is determined from the circuit when the switch is closed. Using Kirchhoff's voltage law around the path of the closed switch, C_1 , and L_2 with the voltage across C_1 assumed to be a constant V_s , gives

$$v_{L2} = v_{C1} = V_s = L_2 \left(\frac{di_{L2}}{dt} \right) = L_2 \left(\frac{\Delta i_{L2}}{\Delta t} \right) = L_2 \left(\frac{\Delta i_{L2}}{DT} \right)$$

Solving for Δi_{L2}

$$\Delta i_{L2} = \frac{V_s DT}{L_2} = \frac{V_s D}{L_2 f} \quad (6-80)$$

Applications of Kirchhoff's current law show that the diode and switch currents are

$$\begin{aligned} i_D &= \begin{cases} 0 & \text{when switch is closed} \\ i_{L1} + i_{L2} & \text{when switch is open} \end{cases} \\ i_{sw} &= \begin{cases} i_{L1} + i_{L2} & \text{when switch is closed} \\ 0 & \text{when switch is open} \end{cases} \end{aligned} \quad (6-81)$$

Current waveforms are shown in Fig. 6-15.

Kirchhoff's voltage law applied to the circuit of Fig. 6-14c, assuming no voltage ripple across the capacitors, shows that the voltage across the switch when it is open is $V_s + V_o$. From Fig. 6-14b, the maximum reverse bias voltage across the diode when it is off is also $V_s + V_o$.

The output stage consisting of the diode, C_2 , and the load resistor is the same as in the boost converter, so the output ripple voltage is

$$\Delta V_o = \Delta V_{C2} = \frac{V_o D}{RC_2 f} \quad (6-82)$$

Solving for C_2 ,

$$C_2 = \frac{D}{R(\Delta V_o/V_o)f} \quad (6-83)$$

The voltage variation in C_1 is determined from the circuit with the switch closed (Fig. 6-14b). Capacitor current i_{C1} is the opposite of i_{L2} , which has previously been determined to have an average value of I_o . From the definition of capacitance and considering the magnitude of charge,

$$\Delta V_{C1} = \frac{\Delta Q_{C1}}{C} = \frac{I_o \Delta t}{C} = \frac{I_o DT}{C}$$

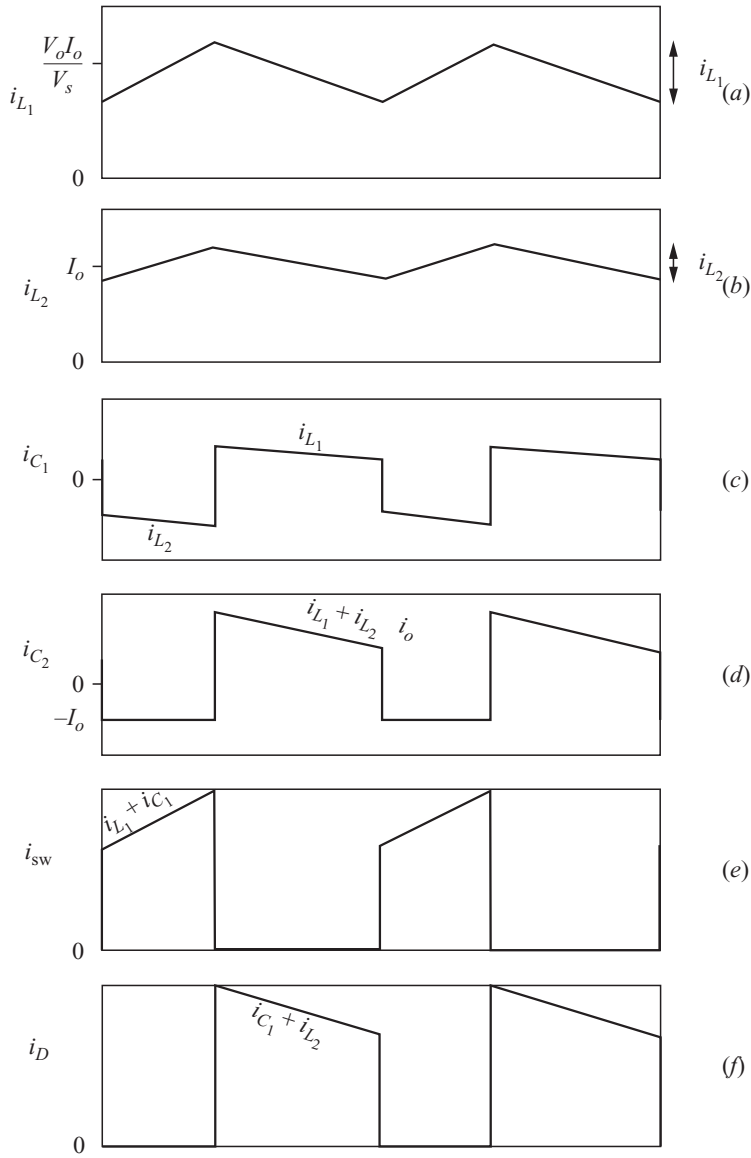


Figure 6-15 Currents in the SEPIC converter. (a) L_1 ; (b) L_2 ; (c) C_1 ; (d) C_2 ; (e) switch; (f) diode.

Replacing I_o with V_o/R ,

$$\Delta V_{C_1} = \frac{V_o D}{RC_1 f} \quad (6-84)$$

Solving for C_1 ,

$$C_1 = \frac{D}{R(\Delta V_{C_1}/V_o)f} \quad (6-85)$$

The effect of equivalent series resistance of the capacitors on voltage variation is usually significant, and the treatment is the same as with the converters discussed previously.

EXAMPLE 6-8**SEPIC Circuit**

The SEPIC circuit of Fig. 6-14a has the following parameters:

$$\begin{aligned} V_s &= 9 \text{ V} \\ D &= 0.4 \\ f &= 100 \text{ kHz} \\ L_1 &= L_2 = 90 \text{ } \mu\text{H} \\ C_1 &= C_2 = 80 \text{ } \mu\text{F} \\ I_o &= 2 \text{ A} \end{aligned}$$

Determine the output voltage; the average, maximum, and minimum inductor currents; and the variation in voltage across each capacitor.

■ Solution

The output voltage is determined from Eq. (6-74).

$$V_o = V_s \left(\frac{D}{1-D} \right) = 9 \left(\frac{0.4}{1-0.4} \right) = 6 \text{ V}$$

The average current in L_1 is determined from Eq. (6-76).

$$I_{L1} = \frac{V_o I_o}{V_s} = \frac{6(2)}{9} = 1.33 \text{ A}$$

From Eq. (6-78)

$$\Delta i_{L1} = \frac{V_s D}{L_1 f} = \frac{9(0.4)}{90(10)^{-6}(100,000)} = 0.4 \text{ A}$$

Maximum and minimum currents in L_1 are then

$$I_{L1,\max} = I_{L1} + \frac{\Delta i_{L1}}{2} = 1.33 + \frac{0.4}{2} = 1.53 \text{ A}$$

$$I_{L1,\min} = I_{L1} - \frac{\Delta i_{L1}}{2} = 1.33 - \frac{0.4}{2} = 1.13 \text{ A}$$

For the current in L_2 , the average is the same as the output current $I_o = 2 \text{ A}$. The variation in I_{L2} is determined from Eq. (6-80)

$$\Delta i_{L2} = \frac{V_s D}{L_2 f} = \frac{9(0.4)}{90(10)^{-6}(100,000)} = 0.4 \text{ A}$$

resulting in maximum and minimum current magnitudes of

$$I_{L2,\max} = 2 + \frac{0.4}{2} = 2.2 \text{ A}$$

$$I_{L2,\min} = 2 - \frac{0.4}{2} = 1.8 \text{ A}$$

Using an equivalent load resistance of $6 \text{ V}/2 \text{ A} = 3 \Omega$, the ripple voltages in the capacitors are determined from Eqs. (6-82) and (6-84).

$$\Delta V_o = \Delta V_{C_2} = \frac{V_o D}{RC_2 f} = \frac{6(0.4)}{(3)80(10)^{-6}(100,000)} = 0.1 \text{ V}$$

$$\Delta V_{C_1} = \frac{V_o D}{RC_1 f} = \frac{6(0.4)}{(3)80(10)^{-6}(100,000)} = 0.1 \text{ V}$$

In Example 6-8, the values of L_1 and L_2 are equal, which is not a requirement. However, when they are equal, the rates of change in the inductor currents are identical [Eqs. (6-78) and (6-80)]. The two inductors may then be wound on the same core, making a 1:1 transformer. Figure 6-16 shows an alternative representation of the SEPIC converter.

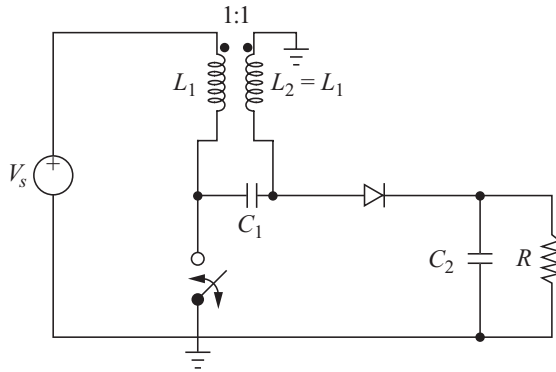


Figure 6-16 A SEPIC circuit using mutually coupled inductors.

6.9 INTERLEAVED CONVERTERS

Interleaving, also called *multiphasing*, is a technique that is useful for reducing the size of filter components. An interleaved buck converter is shown in Fig. 6-17a. This is equivalent to a parallel combination of two sets of switches, diodes, and inductors connected to a common filter capacitor and load. The switches are operated 180° out of phase, producing inductor currents that are also 180° out of phase. The current entering the capacitor and load resistance is the sum of the inductor currents, which has a smaller peak-to-peak variation and a frequency twice as large as individual inductor currents. This results in a smaller peak-to-peak variation in capacitor current than would be achieved with a single buck converter, requiring less capacitance for the same output ripple voltage. The variation in current coming from the source is also reduced. Figure 6-17b shows the current waveforms.

The output voltage is obtained by taking Kirchhoff's voltage law around either path containing the voltage source, a switch, an inductor, and the output voltage. The voltage across the inductor is $V_s - V_o$ with the switch closed and

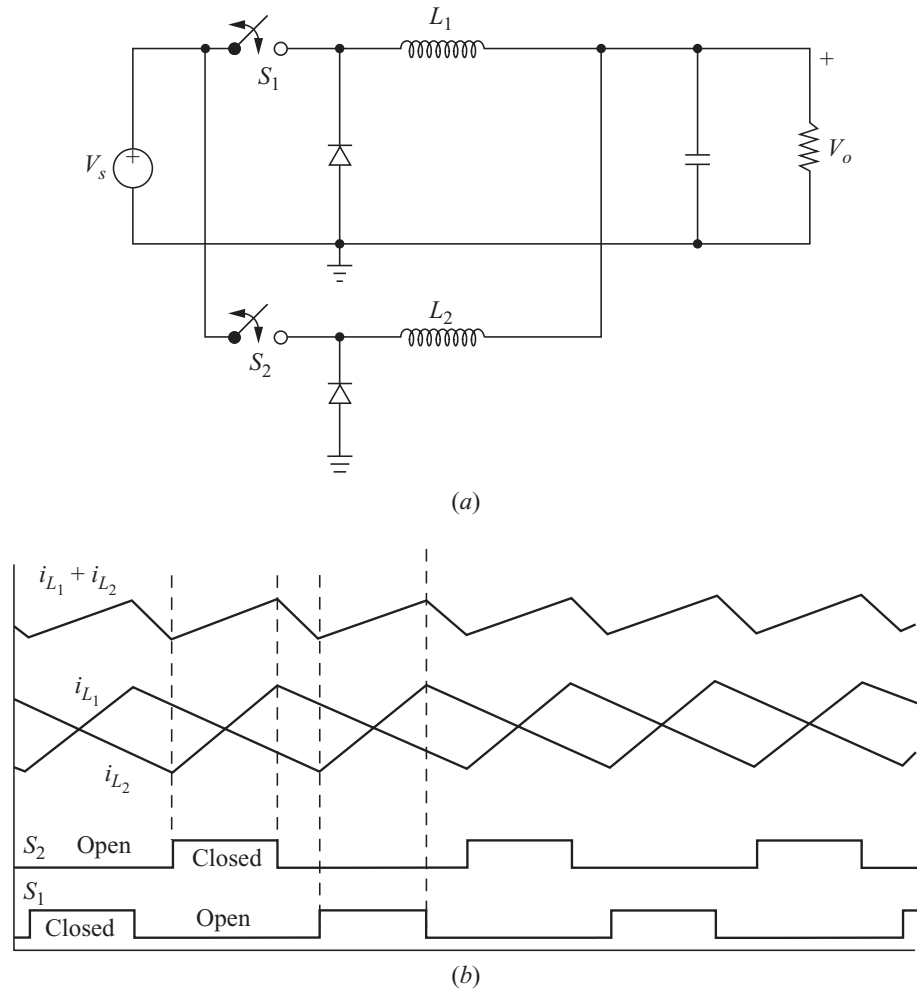


Figure 6-17 (a) An interleaved buck converter; (b) The switching scheme and current waveforms.

is $-V_o$ with the switch open. These are the same as for the buck converter of Fig. 6-3a discussed previously, resulting in

$$V_o = V_s D$$

where D is the duty ratio of each switch.

Each inductor supplies one-half of the load current and output power, so the average inductor current is one-half of what it would be for a single buck converter.

More than two converters can be interleaved. The phase shift between switch closing is $360^\circ/n$, where n is the number of converters in the parallel configuration. Interleaving can be done with the other converters in this chapter and with the converters that are described in Chap. 7. Figure 6-18 shows an interleaved boost converter.

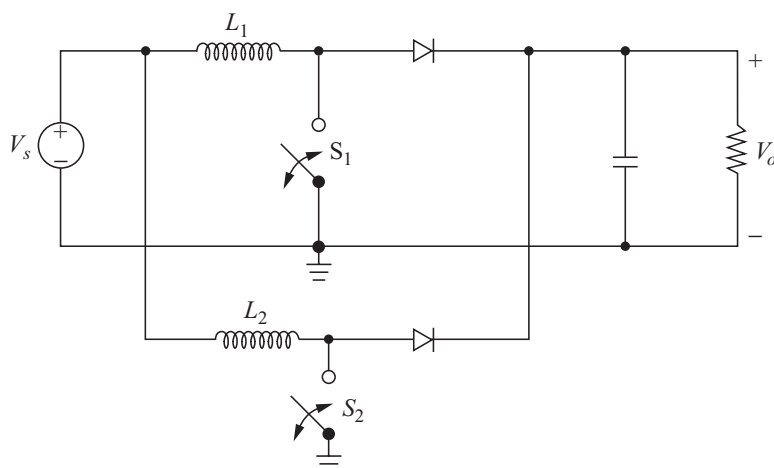


Figure 6-18 An interleaved boost converter.

6.10 NONIDEAL SWITCHES AND CONVERTER PERFORMANCE

Switch Voltage Drops

All the preceding calculations were made with the assumption that the switches were ideal. Voltage drops across conducting transistors and diodes may have a significant effect on converter performance, particularly when the input and output voltages are low. Design of dc-dc converters must account for nonideal components. The buck converter is used to illustrate the effects of switch voltage drops.

Referring again to the analysis of the buck converter of Fig. 6-3a, the input-output voltage relationship was determined using the inductor voltage and current. With nonzero voltage drops across conducting switches, the voltage across the inductor with the switch closed becomes

$$v_L = V_s - V_o - V_Q \quad (6-86)$$

where V_Q is the voltage across the conducting switch. With the switch open, the voltage across the diode is V_D and the voltage across the inductor is

$$v_L = -V_o - V_D \quad (6-87)$$

The average voltage across the inductor is zero for the switching period.

$$V_L = (V_s - V_o - V_Q)D + (-V_o - V_D)(1 - D) = 0$$

Solving for V_o ,

$$V_o = V_s D - V_Q D - V_D(1 - D) \quad (6-88)$$

which is lower than $V_o = V_s D$ for the ideal case.

Switching Losses

In addition to the on-state voltage drops and associated power losses of the switches, other losses occur in the switches as they turn on and off. Figure 6-19a illustrates switch on-off transitions. For this case, it is assumed that the changes in voltage and current are linear and that the timing sequence is as shown. The instantaneous power dissipated in the switch is shown in Fig. 6-19a. Another possible switch on-off transition is shown in Fig. 6-12b. In this case, the voltage and current transitions do not occur simultaneously. This may be closer to actual switching situations, and switching power loss is larger for this case. (See Chap. 10 for additional information.)

The energy loss in one switching transition is the area under the power curve. Since the average power is energy divided by the period, higher switching frequencies result in higher switching losses. One way to reduce switching losses is to modify the circuit to make switching occur at zero voltage and/or zero current. This is the approach of the resonant converter, which is discussed in Chap. 9.

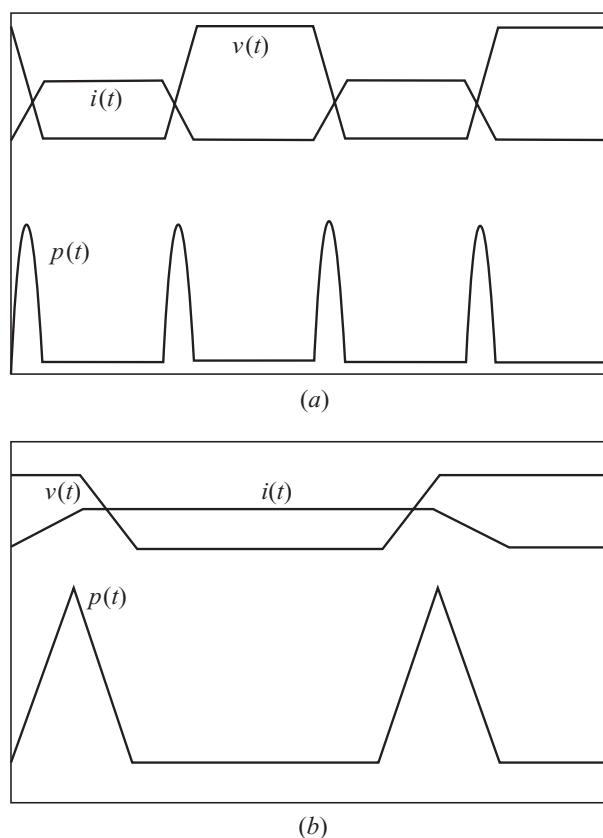


Figure 6-19 Switch voltage, current, and instantaneous power. (a) Simultaneous voltage and current transition; (b) Worst-case transition.

6.11 DISCONTINUOUS-CURRENT OPERATION

Continuous current in the inductor was an important assumption in the previous analyses for dc-dc converters. Recall that continuous current means that the current in the inductor remains positive for the entire switching period. Continuous current is not a necessary condition for a converter to operate, but a different analysis is required for the discontinuous-current case.

Buck Converter with Discontinuous Current

Figure 6-20 shows the inductor and source currents for discontinuous-current operation for the buck converter of Fig 6-3a. The relationship between output and input voltages is determined by first recognizing that the average inductor voltage is zero for periodic operation. From the inductor voltage shown in Fig. 6-20c,

$$(V_s - V_o)DT - V_o D_1 T = 0$$

which is rearranged to get

$$(V_s - V_o)D = V_o D_1 \quad (6-89)$$

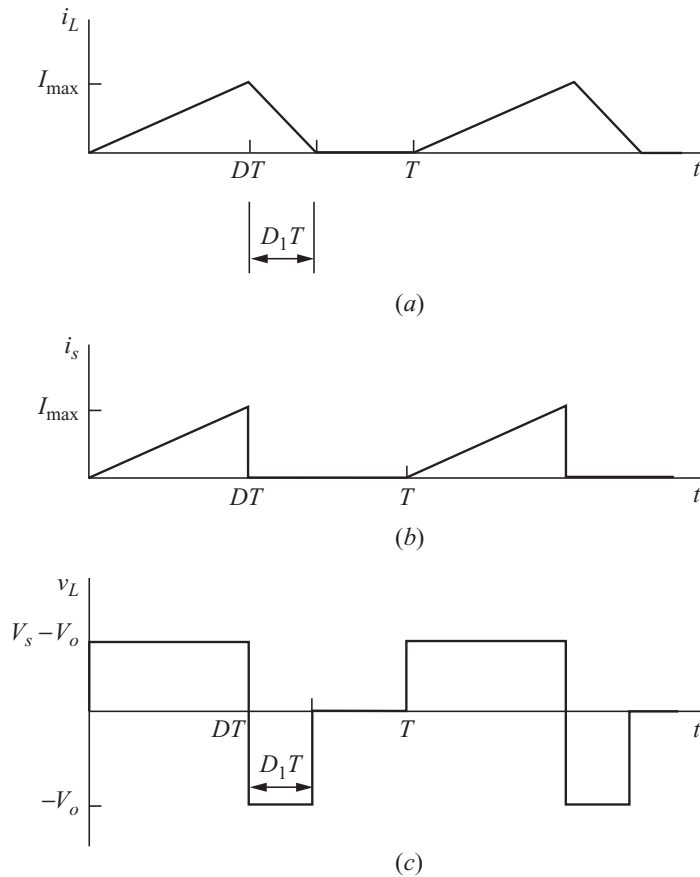


Figure 6-20 Buck converter discontinuous current.
(a) Inductor current; (b) Source current; (c) Inductor voltage.

$$\frac{V_o}{V_s} = \left(\frac{D}{D + D_1} \right) \quad (6-90)$$

Next, the average inductor current equals the average resistor current because the average capacitor current is zero. With the output voltage assumed constant,

$$I_L = I_R = \frac{V_o}{R}$$

Computing the average inductor current from Fig. 6-20a,

$$I_L = \frac{1}{T} \left(\frac{1}{2} I_{\max} DT + \frac{1}{2} I_{\max} D_1 T \right) = \frac{1}{2} I_{\max} (D + D_1)$$

which results in

$$\frac{1}{2} I_{\max} (D + D_1) = \frac{V_o}{R} \quad (6-91)$$

Since the current starts at zero, the maximum current is the same as the change in current over the time that the switch is closed. With the switch closed, the voltage across the inductor is

$$v_L = V_s - V_o$$

which results in

$$\frac{di_L}{dt} = \frac{V_s - V_o}{L} = \frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{I_{\max}}{DT} \quad (6-92)$$

Solving for $I_{\max}$ and using Eq. (6-89) for $(V_s - V_o)D$,

$$I_{\max} = \Delta i_L = \left(\frac{V_s - V_o}{L} \right) DT = \frac{V_o D_1 T}{L} \quad (6-93)$$

Substituting for $I_{\max}$ in Eq. (6-91),

$$\frac{1}{2} I_{\max} (D + D_1) = \frac{1}{2} \left(\frac{V_o D_1 T}{L} \right) (D + D_1) = \frac{V_o}{R} \quad (6-94)$$

which gives

$$D_1^2 + DD_1 - \frac{2L}{RT} = 0$$

Solving for D_1 ,

$$D_1 = \frac{-D + \sqrt{D^2 + 8L/RT}}{2} \quad (6-95)$$

Substituting for D_1 in Eq. (6-90),

$$V_o = V_s \left(\frac{D}{D + D_1} \right) = V_s \left[\frac{2D}{D + \sqrt{D^2 + 8L/RT}} \right] \quad (6-96)$$

The boundary between continuous and discontinuous current occurs when $D_1 = 1 - D$. Recall that another condition that occurs at the boundary between continuous and discontinuous current is $I_{\min} = 0$ in Eq. (6-12).

EXAMPLE 6-9

Buck Converter with Discontinuous Current

For the buck converter of Fig. 6-3a,

$$\begin{aligned} V_s &= 24 \text{ V} \\ L &= 200 \text{ } \mu\text{H} \\ R &= 20 \text{ } \Omega \\ C &= 1000 \text{ } \mu\text{F} \\ f &= 10 \text{ kHz} \quad \text{switching frequency} \\ D &= 0.4 \end{aligned}$$

(a) Show that the inductor current is discontinuous, (b) Determine the output voltage V_o .

■ Solution

(a) For discontinuous current, $D_1 < 1 - D$, and D_1 is calculated from Eq. (6-95).

$$\begin{aligned} D_1 &= \frac{-D + \sqrt{D^2 + 8L/RT}}{2} \\ &= \frac{1}{2} \left(-0.4 + \sqrt{0.4^2 + \frac{8(200)(10)^{-6}(10,000)}{20}} \right) = 0.29 \end{aligned}$$

Comparing D_1 to $1 - D$, $0.29 < (1 - 0.4)$ shows that the inductor current is discontinuous. Alternatively, the minimum inductor current computed from Eq. (6-12) is $I_{\min} = -0.96 \text{ A}$. Since negative inductor current is not possible, inductor current must be discontinuous.

(b) Since D_1 is calculated and discontinuous current is verified, the output voltage can be computed from Eq. (6-96).

$$V_o = V_s \left(\frac{D}{D + D_1} \right) = 20 \left(\frac{0.4}{0.4 + 0.29} \right) = 13.9 \text{ V}$$

Figure 6-21 shows the relationship between output voltage and duty ratio for the buck converter of Example 6-9. All parameters except D are those of Example 6-9. Note the linear relationship between input and output for continuous current and the nonlinear relationship for discontinuous current. For a given duty ratio, the output voltage is greater for discontinuous-current operation than it would be if current were continuous.

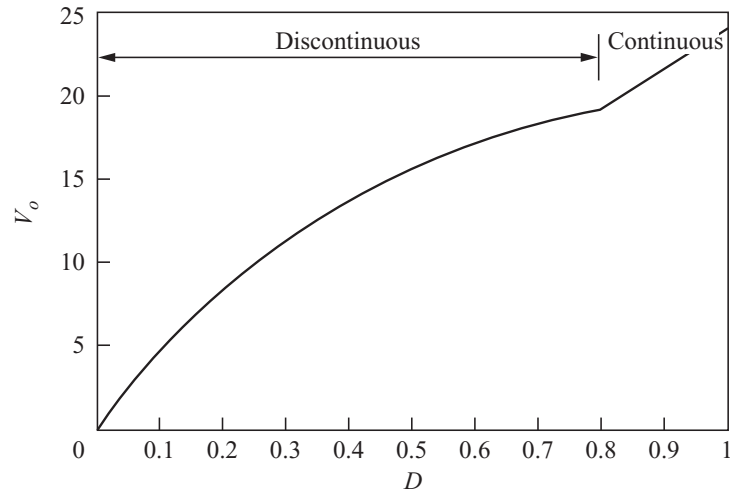


Figure 6-21 V_o versus duty ratio for the buck converter of Example 6-9.

Boost Converter with Discontinuous Current

The boost converter will also operate for discontinuous inductor current. In some cases, the discontinuous-current mode is desirable for control reasons in the case of a regulated output. The relationship between output and input voltages is determined from two relationships:

1. The average inductor voltage is zero.
2. The average current in the diode is the same as the load current.

The inductor and diode currents for discontinuous current have the basic waveforms as shown in Fig. 6-22*a* and *c*. When the switch is on, the voltage across the inductor is V_s . When the switch is off and the inductor current is positive, the inductor voltage is $V_s - V_o$. The inductor current decreases until it reaches zero and is prevented from going negative by the diode. With the switch open and the diode off, the inductor current is zero. The average voltage across the inductor is

$$V_s DT + (V_s - V_o) D_1 T = 0$$

which results in

$$V_o = V_s \left(\frac{D + D_1}{D_1} \right) \quad (6-97)$$

The average diode current (Fig. 6-22*c*) is

$$I_D = \frac{1}{T} \left(\frac{1}{2} I_{\max} D_1 T \right) = \frac{1}{2} I_{\max} D_1 \quad (6-98)$$

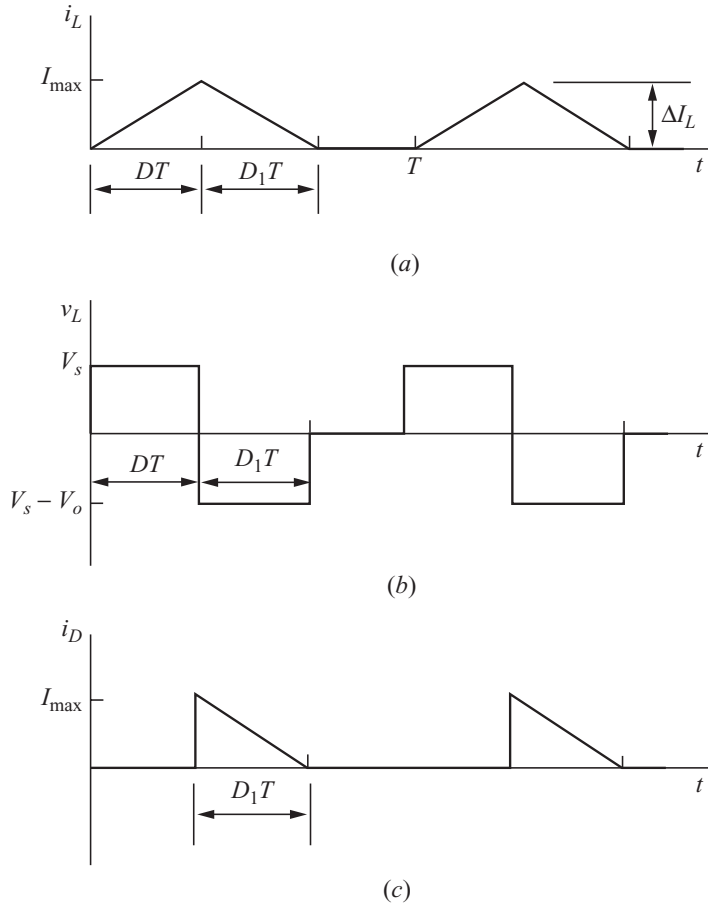


Figure 6-22 Discontinuous current in the boost converter.
 (a) Inductor current; (b) Inductor voltage; (c) Diode current.

Current $I_{\max}$ is the same as the change in inductor current when the switch is closed.

$$I_{\max} = \Delta i_L = \frac{V_s DT}{L} \quad (6-99)$$

Substituting for $I_{\max}$ in Eq. (6-98) and setting the result equal to the load current,

$$I_D = \frac{1}{2} \left(\frac{V_s DT}{L} \right) D_1 = \frac{V_o}{R} \quad (6-100)$$

Solving for D_1 ,

$$D_1 = \left(\frac{V_o}{V_s} \right) \left(\frac{2L}{RDT} \right) \quad (6-101)$$

Substituting the preceding expression for D_1 into Eq. (6-97) results in the quadratic equation

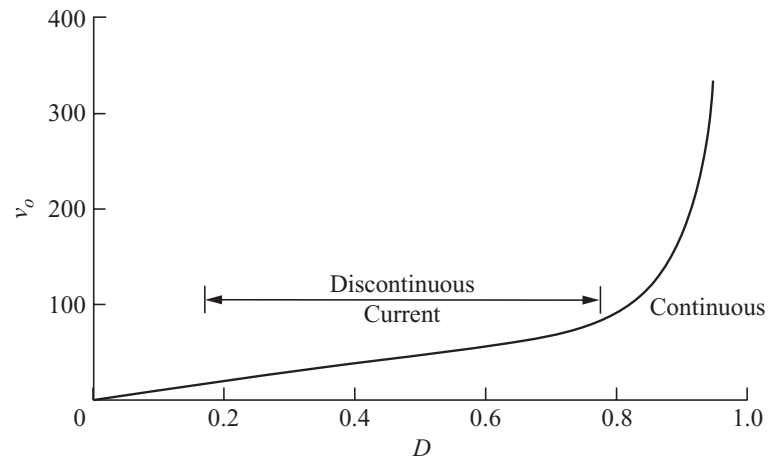


Figure 6-23 Output voltage of boost converter.

$$\left(\frac{V_o}{V_s}\right)^2 - \frac{V_o}{V_s} - \frac{D^2 RT}{2L} = 0$$

Solving for V_o/V_s ,

$$\frac{V_o}{V_s} = \frac{1}{2} \left(1 + \sqrt{1 + \frac{2D^2 RT}{L}} \right) \quad (6-102)$$

The boundary between continuous and discontinuous current occurs when $D_1 = 1 - D$. Another condition at the boundary is when $I_{\min}$ in Eq. (6-30) is zero.

Whether the boost converter is operating in the continuous or discontinuous mode depends on the combination of circuit parameters, including the duty ratio. As the duty ratio for a given boost converter is varied, the converter may go into and out of the discontinuous mode. Figure 6-23 shows the output voltage for a boost converter as the duty ratio is varied.

EXAMPLE 6-10

Boost Converter with Discontinuous Current

The boost converter of Fig. 6-8a has parameters

$$\begin{aligned} V_s &= 20 \text{ V} \\ D &= 0.6 \\ L &= 100 \text{ } \mu\text{H} \\ R &= 50 \text{ } \Omega \\ C &= 100 \text{ } \mu\text{F} \\ f &= 15 \text{ kHz} \end{aligned}$$

- (a) Verify that the inductor current is discontinuous, (b) determine the output voltage, and (c) determine the maximum inductor current.

■ Solution

- (a) First assume that the inductor current is continuous and compute the minimum from Eq. (6-30), resulting in $I_{\min} = -1.5$ A. Negative inductor current is not possible, indicating discontinuous current.
- (b) Equation (6-102) gives the output voltage

$$V_o = \frac{V_s}{2} \left(1 + \sqrt{1 + \frac{2D^2 R}{Lf}} \right) = \frac{20}{2} \left[1 + \sqrt{1 + \frac{2(0.6)^2(50)}{100(10)^{-6}(15,000)}} \right] = 60 \text{ V}$$

Note that a boost converter with the same duty ratio operating with continuous current would have an output of 50 V.

- (c) The maximum inductor current is determined from Eq. (6-99).

$$I_{\max} = \frac{V_s D}{Lf} = \frac{(20)(0.6)}{100(10)^{-6}(15,000)} = 8 \text{ A}$$

6.12 SWITCHED-CAPACITOR CONVERTERS

In switched-capacitor converters, capacitors are charged in one circuit configuration and then reconnected in a different configuration, producing an output voltage different from the input. Switched-capacitor converters do not require an inductor and are also known as *inductorless converters* or *charge pumps*. Switched-capacitor converters are useful for applications that require small currents, usually less than 100 mA. Applications include use in RS-232 data signals that require both positive and negative voltages for logic levels; in flash memory circuits, where large voltages are needed to erase stored information; and in drivers for LEDs and LCD displays.

The basic types of switched-capacitor converters are the step-up (boost), the inverting, and the step-down (buck) circuits. The following discussion introduces the concepts of switched-capacitor converters.

The Step-Up Switched-Capacitor Converter

A common application of a switched-capacitor converter is the step-up (boost) converter. The basic principle is shown in Fig. 6-24a. A capacitor is first connected across the source to charge it to V_s . The charged capacitor is then connected in series with the source, producing an output voltage of $2V_s$.

A switching scheme to accomplish this is shown in Fig. 6-24b. The switch pair labeled 1 is closed and opened in a phase sequence opposite to that of switch pair 2. Switch pair 1 closes to charge the capacitor and then opens. Switch pair 2 then closes to produce an output of $2V_s$.

The switches can be implemented with transistors, or they can be implemented with transistors and diodes, as shown in Fig. 6-24c. Transistor M_1 is turned on, and C_1 is charged to V_s through D_1 . Next, M_1 is turned off and M_2 is turned on. Kirchhoff's voltage law around the path of the source, the charged

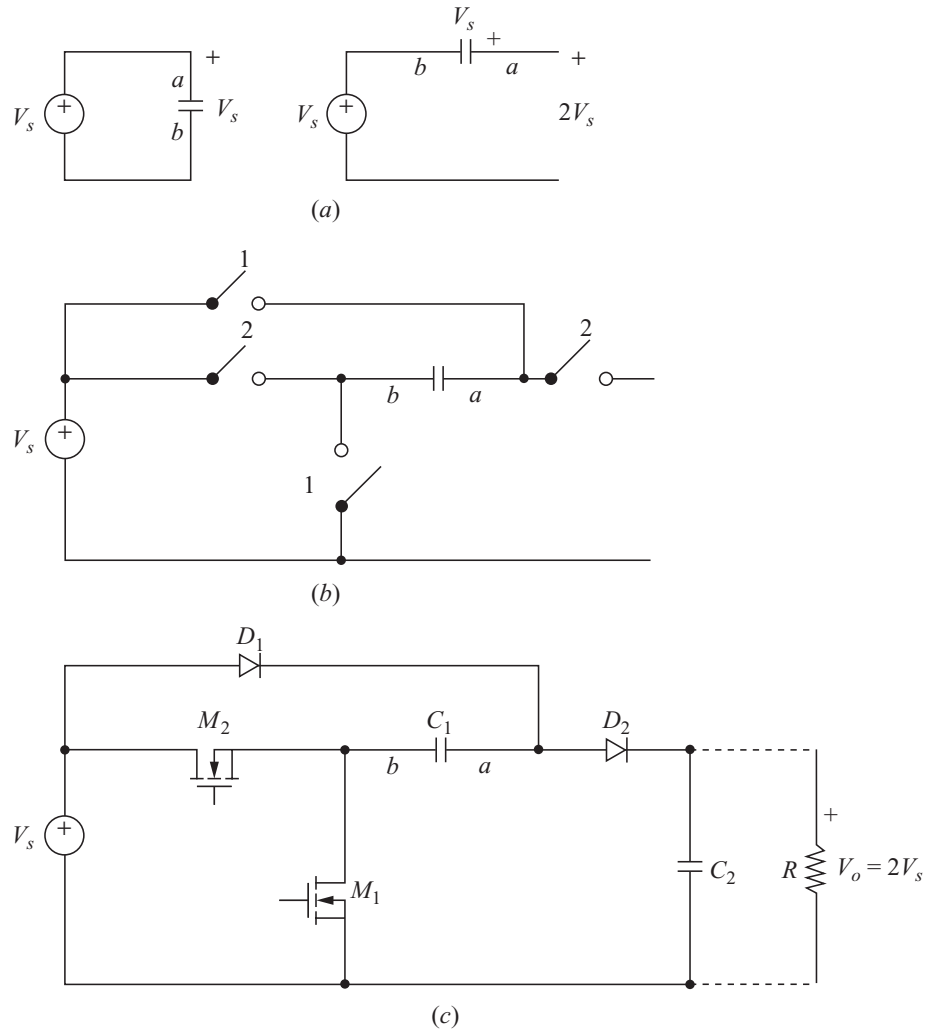


Figure 6-24 A switched-capacitor step-up converter. (a) A capacitor is charged and then reconnected to produce a voltage of twice that of the source; (b) A switch arrangement; (c) An implementation using transistors and diodes and showing a second capacitor C_2 to sustain the output voltage during switching.

capacitor C_1 , and V_o shows that $V_o = 2V_s$. The capacitor C_2 on the output is required to sustain the output voltage and to supply load current when C_1 is disconnected from the load. With C_2 included, it will take several switching cycles to charge it and achieve the final output voltage. With the resistor connected, current will flow from the capacitors, but the output voltage will be largely unaffected if the switching frequency is sufficiently high and capacitor charges are replenished in short time intervals. The output will be less than $2V_s$ for real devices because of voltage drops in the circuit.

Converters can be made to step up the input voltage to values greater than $2V_s$. In Fig. 6-25a, two capacitors are charged and then reconnected to create a

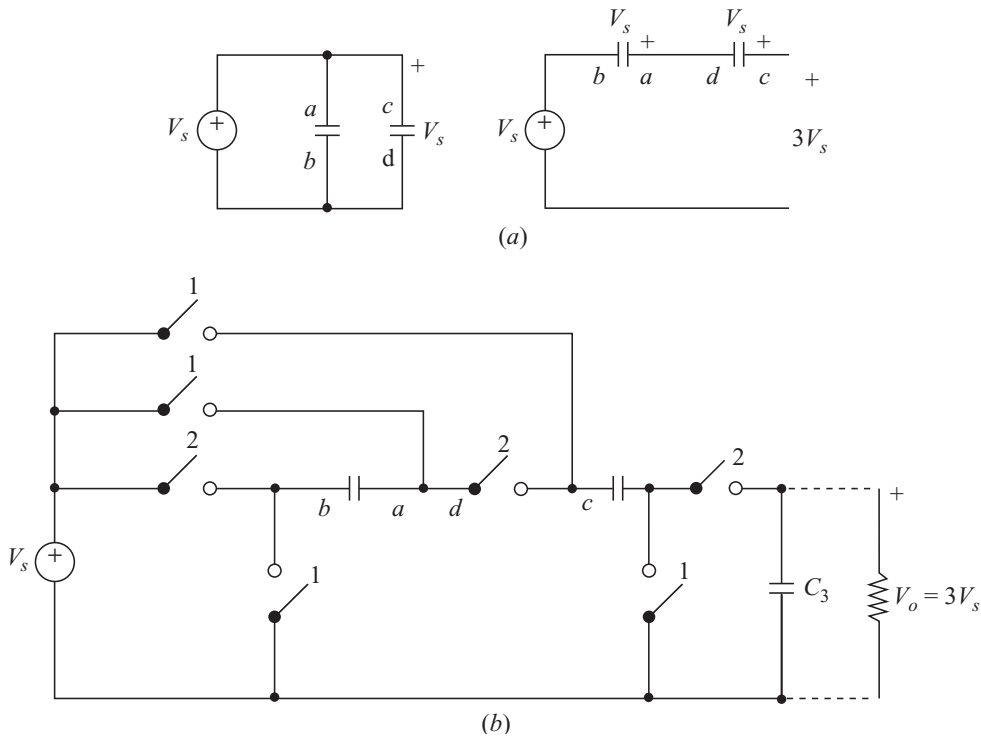


Figure 6-25 A step-up switched-capacitor converter to produce 3 times the source voltage. (a) Each capacitor is charged to V_s and reconnected to produce an output of $3V_s$; (b) A switch arrangement also shows an output capacitor to sustain the output voltage during switching.

voltage of $3V_s$. A switching arrangement to implement this circuit is shown in Fig. 6-25b. Switch sets 1 and 2 open and close alternately. The circuit includes an output capacitor C_3 to sustain the voltage across the load during the switching cycle.

The Inverting Switched-Capacitor Converter

The inverting switched-capacitor converter is useful for producing a negative voltage from a single voltage source. For example, -5 V can be made from a 5-V source, thereby creating a ± 5 -V supply. The basic concept is shown in Fig. 6-26a. A capacitor is charged to the source voltage and then connected to the output with opposite polarity.

A switching scheme to accomplish this is shown in Fig. 6-26b. Switch pairs 1 and 2 open and close in opposite phase sequence. Switch pair 1 closes to charge the capacitor and then opens. Switch pair 2 then closes to produce an output of $-V_s$.

A switch configuration to implement the inverting circuit is shown in Fig. 6-26c. An output capacitor C_2 is included to sustain the output and supply current to the load during the switching cycle. Transistor M_1 is turned on, charging C_1 to V_s through D_1 . Transistor M_1 is turned off and M_2 is turned on, charging C_2 with a

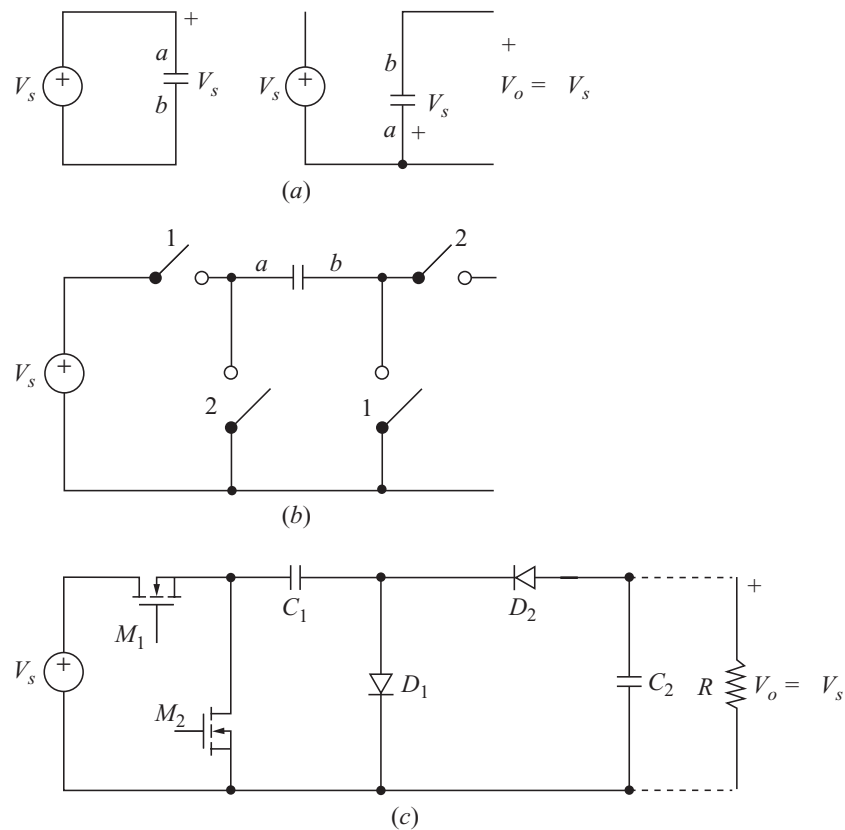


Figure 6-26 The inverting switched-capacitor converter. (a) The capacitor is charged to V_s and then reconnected to produce an output of $-V_s$; (b) A switch arrangement; (c) An implementation using transistors and diodes and showing a second capacitor to sustain the output voltage during switching.

polarity that is positive on the bottom. After several switching cycles, the output voltage is $-V_s$.

The Step-Down Switched-Capacitor Converter

A step-down (buck) switched-capacitor converter is shown in Fig. 6-27. In Fig. 6-27a, two capacitors of equal value are connected in series, resulting in a voltage of $V_s/2$ across each. The capacitors are then reconnected in parallel, making the output voltage $V_s/2$. A switching scheme to accomplish this is shown in Fig. 6-27b. Switch pairs 1 and 2 open and close in opposite phase sequence. With the resistor connected, current will flow from the capacitors, but the output voltage will be unaffected if the switching frequency is sufficiently high and capacitor charges are replenished in short time intervals.

A switch configuration to implement the inverting circuit is shown in Fig. 6-27c. Transistor M_1 is turned on, and both capacitors charge through D_1 .

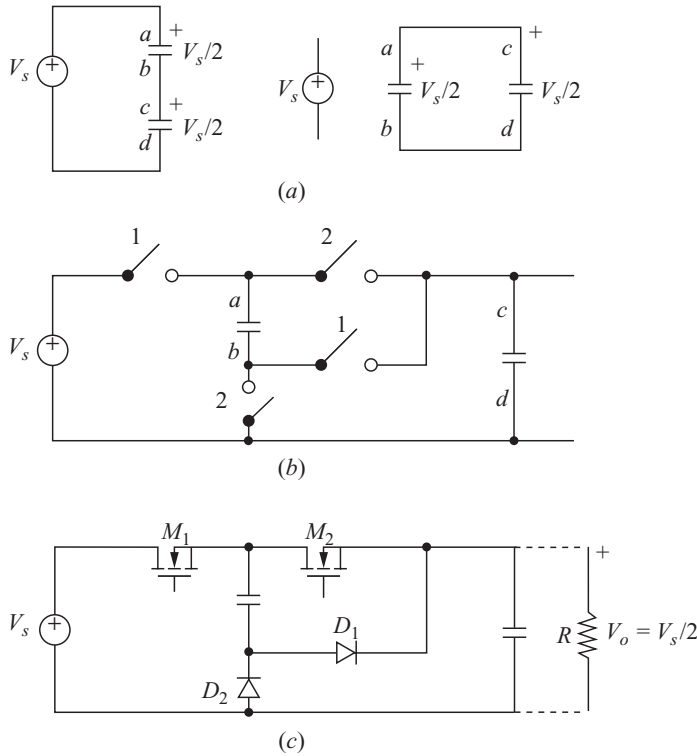


Figure 6-27 The step-down switched-capacitor converter. (a) The capacitors are in series and each is charged to $V_s/2$, followed by the capacitors in parallel, with the output voltage at $V_s/2$; (b) A switch arrangement; (c) An implementation using transistors and diodes.

Transistor M_1 is turned off, and M_2 is turned on, connecting the capacitors in parallel through D_2 . And D_2 is forward-biased as the capacitors discharge into the load resistor.

6.13 PSPIICE SIMULATION OF DC-DC CONVERTERS

The circuit model to be used for PSpice simulation of the dc-dc converters discussed in this chapter depends on the ultimate goal of the simulation. To predict the behavior of a circuit with the goal of producing the periodic voltage and current waveforms requires a circuit model that includes a switch. A voltage-controlled switch is convenient for this application. If the circuit includes an ideal diode and lossless inductors and capacitors, the simulation results will be first-order approximations of circuit behavior, much the same as the analytical work done previously in this chapter. By including parasitic elements and using nonideal switching devices in the circuit model, the simulation will be useful to investigate how a real circuit is expected to depart from the ideal.

Another simulation goal may be to predict the dynamic behavior of a dc-dc converter for changes in the source voltage or load current. A disadvantage of

using the cycle-to-cycle switched model is that the time for overall circuit transients may be orders of magnitude larger than the switching period, thereby making the program execution time quite long. A circuit model that does not include the cycle-by-cycle details but does simulate the large-scale dynamic behavior by using averaging techniques may be preferred. PSpice simulations for both cycle-to-cycle and large-scale dynamic behavior are discussed in this section.

A Switched PSpice Model

A voltage-controlled switch is a simple way to model a transistor switch that would actually be used in a physical converter. The voltage-controlled switch has an on resistance that could be selected to match the transistor's, or the on resistance could be chosen negligibly small to simulate an ideal switch. A pulse voltage source acts as the control for the switch.

When periodic closing and opening of the switch in a dc-dc converter begins, a transient response precedes the steady-state voltages and currents described earlier in this chapter. The following example illustrates a PSpice simulation for a buck converter using idealized models for circuit components.

EXAMPLE 6-11

Buck Converter Simulation Using Idealized Components

Use PSpice to verify the buck converter design in Example 6-3.

The buck converter has the following parameters:

$$\begin{aligned} V_s &= 3.3 \text{ V} \\ L &= 1 \text{ } \mu\text{H} \\ C &= 667 \text{ } \mu\text{F} && \text{with an ESR of } 15 \text{ m}\Omega \\ R &= 0.3 \text{ } \Omega && \text{for a load current of } 4 \text{ A} \\ D &= 0.364 && \text{for an output of } 1.2 \text{ V} \\ \text{Switching frequency} &= 500 \text{ kHz} \end{aligned}$$

■ Solution

A PSpice model for the buck converter is shown in Fig. 6-28. A voltage-controlled switch (Sbreak) is used for the switching transistor, with the on resistance R_{on} set to $1 \text{ m}\Omega$ to approximate an ideal device. An ideal diode is simulated by letting the diode parameter n (the emission coefficient in the diode equation) be 0.001. The switch is controlled by a pulse voltage source. The parameter statements file facilitates modification of the circuit file for other buck converters. Initial conditions for the inductor current and capacitor voltage are assumed to be zero to demonstrate the transient behavior of the circuit.

Figure 6-29a shows the Probe output for inductor current and capacitor voltage. Note that there is a transient response of the circuit before the steady-state periodic condition is reached. From the steady-state portion of the Probe output shown in Fig. 6-29b, the maximum and minimum values of the output voltage are 1.213 and 1.1911 V, respectively, for a peak-to-peak variation of about 22 mV, agreeing well with the 24-mV design objective. The maximum and minimum inductor currents are about 4.77 and 3.24 A, agreeing well with the 4.8- and 3.2-A design objectives.

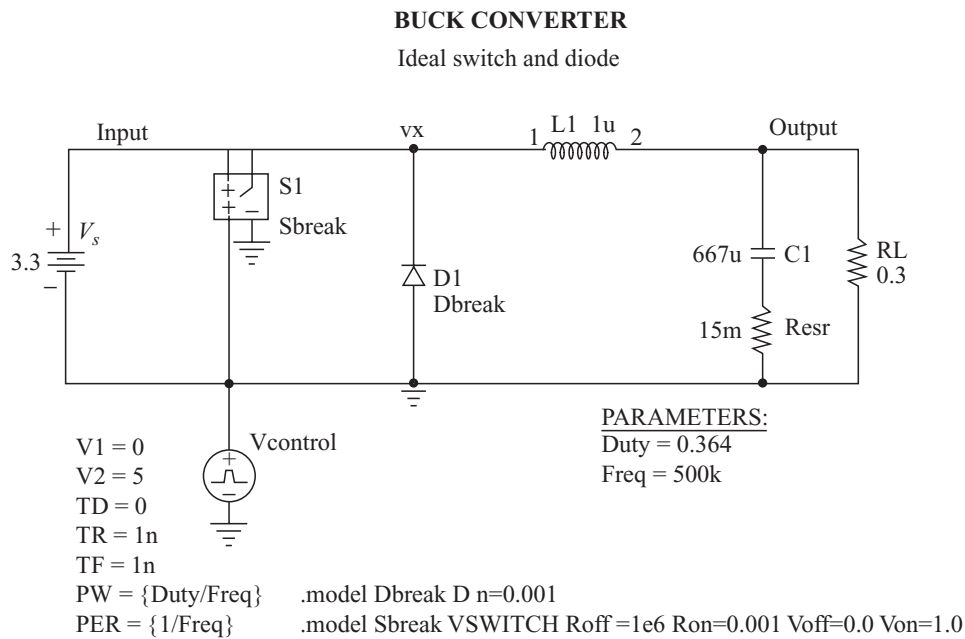


Figure 6-28 PSpice circuit for the buck converter.

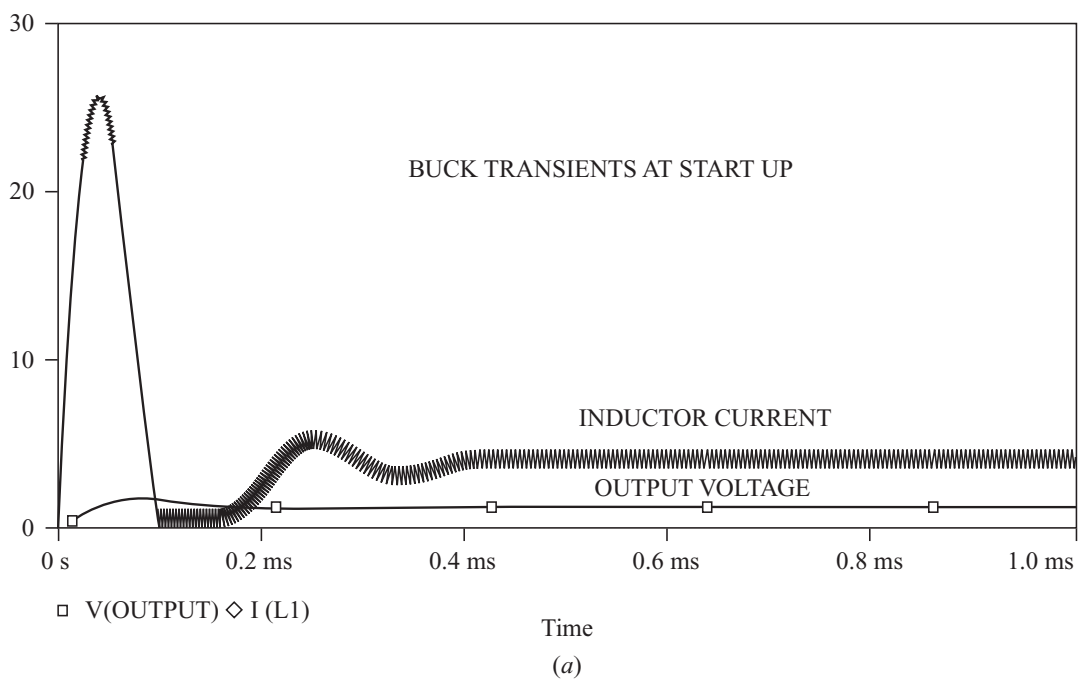


Figure 6-29 Probe output for Example 6-11 (a) showing the transient at start-up and (b) in steady state.

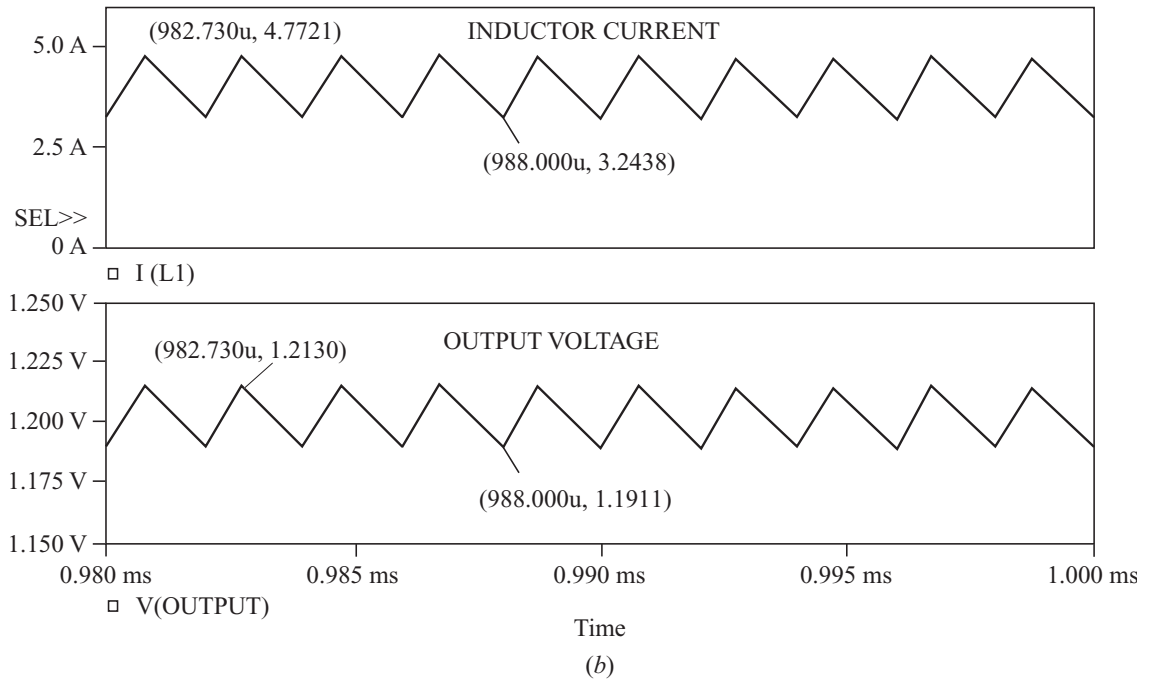


Figure 6-29 (continued)

An Averaged Circuit Model

PSpice simulation of the dc-dc buck converter in Example 6-11 includes both the large scale transient behavior and the cycle-to-cycle waveforms of voltage and current. If the goal of a simulation is to determine the large-scale transient behavior, the cycle-to-cycle response merely adds to the execution time of the program. A more time-efficient way to simulate the transient behavior of dc-dc converters is to use a circuit model that produces the *average* values of voltages and currents only, rather than including the detailed variations around the averages. In general, transient behavior for dc-dc converters can be predicted by analyzing linear networks, with the response equal to the average value of the switching waveforms. The discussion that follows is focused on the buck converter operating in the continuous-current mode.

The transient behavior of the average output voltage can be described using linear circuit analysis. The input v_x to the RLC circuit of the buck converter of Fig. 6-3a has an average value of $V_x = V_s D$. The response of the RLC circuit to a step input voltage of $v_x(t) = (V_s D)u(t)$ represents the average of the output voltage and current waveforms when the converter is turned on. This represents the same large-scale transient that was present in the PSpice simulation shown in Fig. 6-29a.

For complete simulation of the large-scale behavior of a dc-dc converter, it is desirable to include the proper voltage and current relationships between the source and the load. Taking the buck converter as an example, the relationship between average voltage and current at the input and output for continuous inductor current is given by

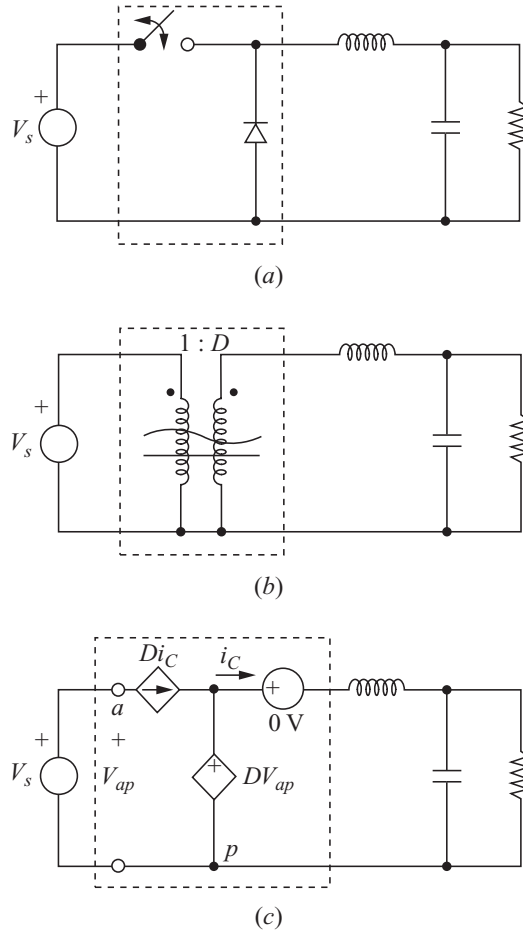


Figure 6-30 (a) Buck converter with switch; (b) Circuit model for averaged buck converter; (c) PSpice circuit.

$$\frac{V_o}{V_s} = \frac{I_s}{I_o} = D \quad (6-103)$$

Since $V_o = V_s D$ and $I_o = I_s/D$, the switch in a model for computing average voltage and current is the same as a “transformer” which has a turns ratio of $1:D$. Circuit models for a buck converter using a $1:D$ transformer and a PSpice circuit for implementing the averaged model are shown in Fig. 6-30. The circuit symbol for the transformer indicates that the model is valid for both ac and dc signals.

The following example illustrates the use of the PSpice model to simulate the response of average voltage and current for a buck converter.

EXAMPLE 6-12

Averaged Buck Converter

Use the averaged circuit of Fig. 6-30c to simulate the buck converter having parameters

$$\begin{aligned}
 V_s &= 10 \text{ V} \\
 D &= 0.2 \\
 L &= 400 \text{ } \mu\text{H} \\
 C &= 400 \text{ } \mu\text{F} \\
 R &= 2 \text{ } \Omega \\
 f &= 5 \text{ kHz}
 \end{aligned}$$

Use initial conditions of zero for inductor current and capacitor voltage.

■ Solution

The PSpice implementation of the averaged model is shown in Fig. 6-31a. The simulation results from both a switched model and for the averaged model are shown in Fig. 6-31b. Note that the switched model shows the cycle-to-cycle variation, while the average model shows only the averaged values.

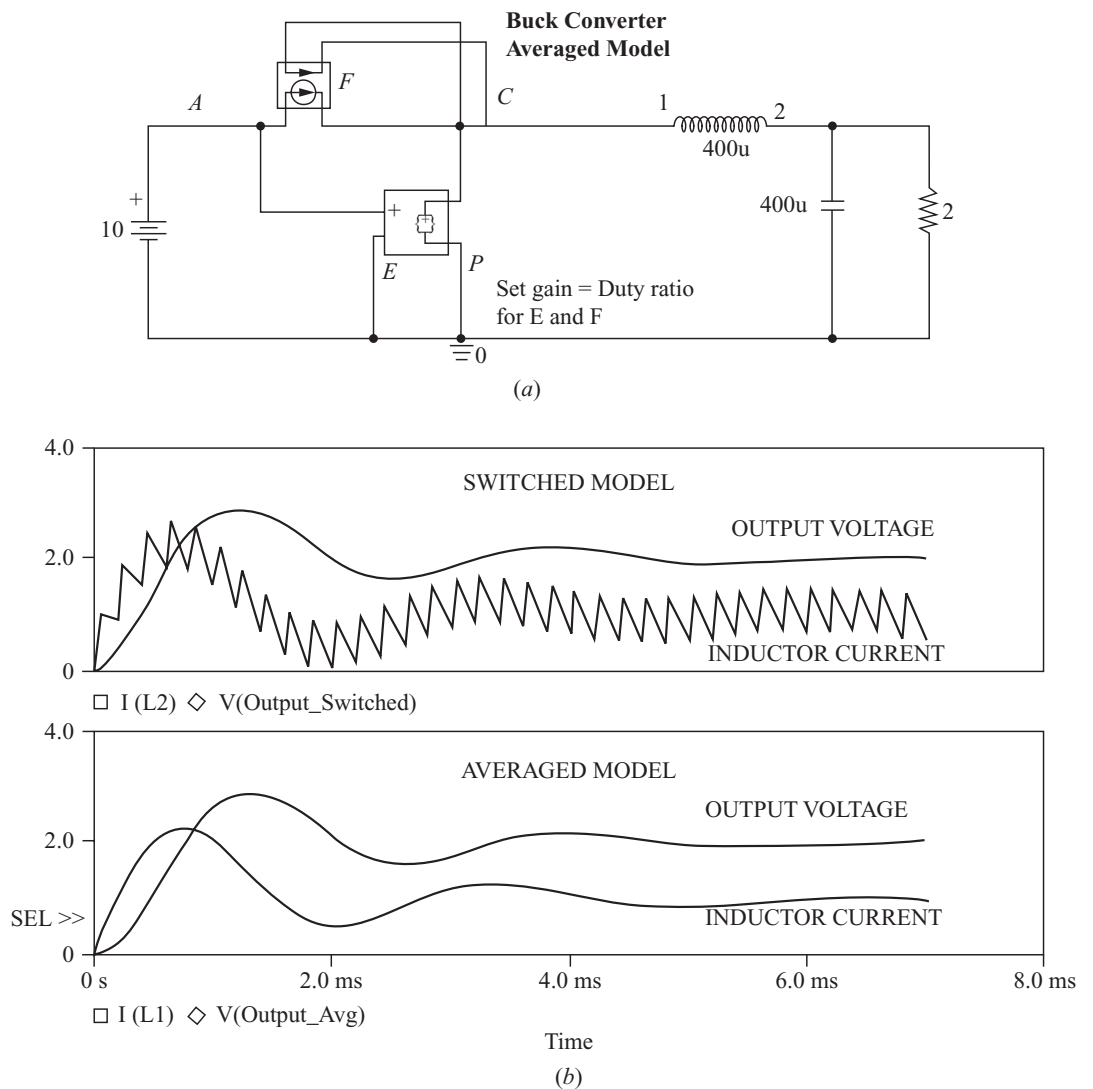


Figure 6-31 (a) PSpice implementation of the averaged buck converter model; (b) Probe output for both the switched model and the averaged model.

The averaged model can be quite useful in investigating the dynamic behavior of the converter when it is subjected to changes in operating parameters. Such an analysis is essential when the output is regulated through a feedback loop which is designed to keep the output at a set level by adjusting the duty ratio of the switch to accommodate variations in the source or the load. Closed-loop response is discussed in Chap. 7 on dc power supplies.

The following example illustrates the use of the averaged circuit model to simulate a step change in load resistance.

EXAMPLE 6-13

Step Change in Load

Use the averaged buck converter model to determine the dynamic response when the load resistance is changed. The circuit parameters are

- $V_s = 50$ V
- $L = 1$ mH with a series resistance of $0.4\ \Omega$
- $C = 100\ \mu\text{F}$ with an equivalent series resistance of $0.5\ \Omega$
- $R = 4\ \Omega$, stepped to $2\ \Omega$ and back to $4\ \Omega$
- $D = 0.4$
- Switching frequency = 5 kHz

■ Solution

Step changes in load are achieved by switching a second $4\text{-}\Omega$ resistor across the output at 6 ms and disconnecting it at 16 ms. The averaged model shows the transients associated with output voltage and inductor current (Fig. 6-32*b*). Also shown for comparison are the results of a different simulation using a switch, showing the cycle-to-cycle variations in voltage and current.

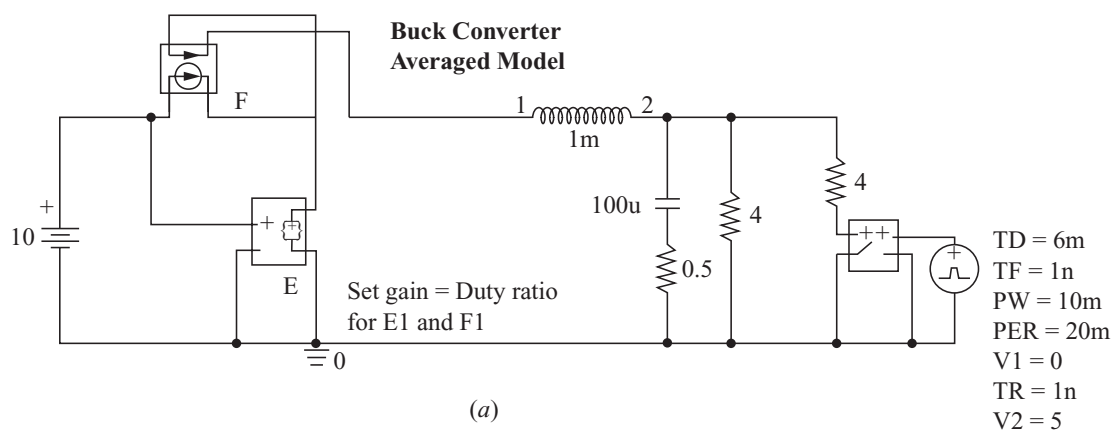


Figure 6-32 (a) PSpice implementation of the averaged model with a switched load; (b) Probe results for both the switched model and the averaged model.

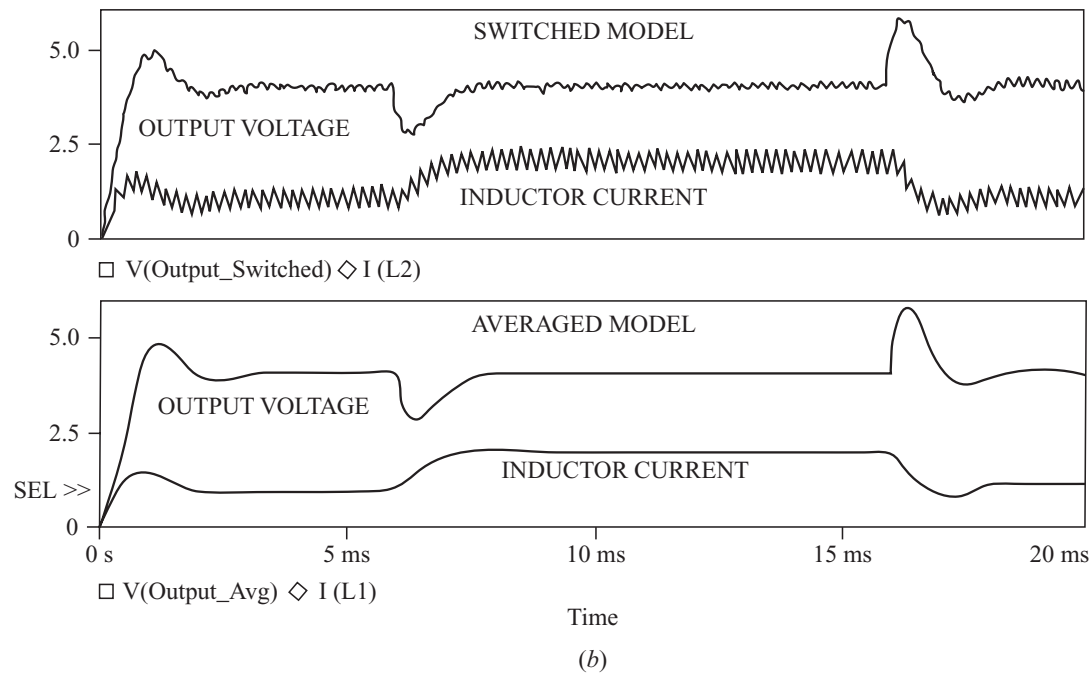


Figure 6-32 (continued)

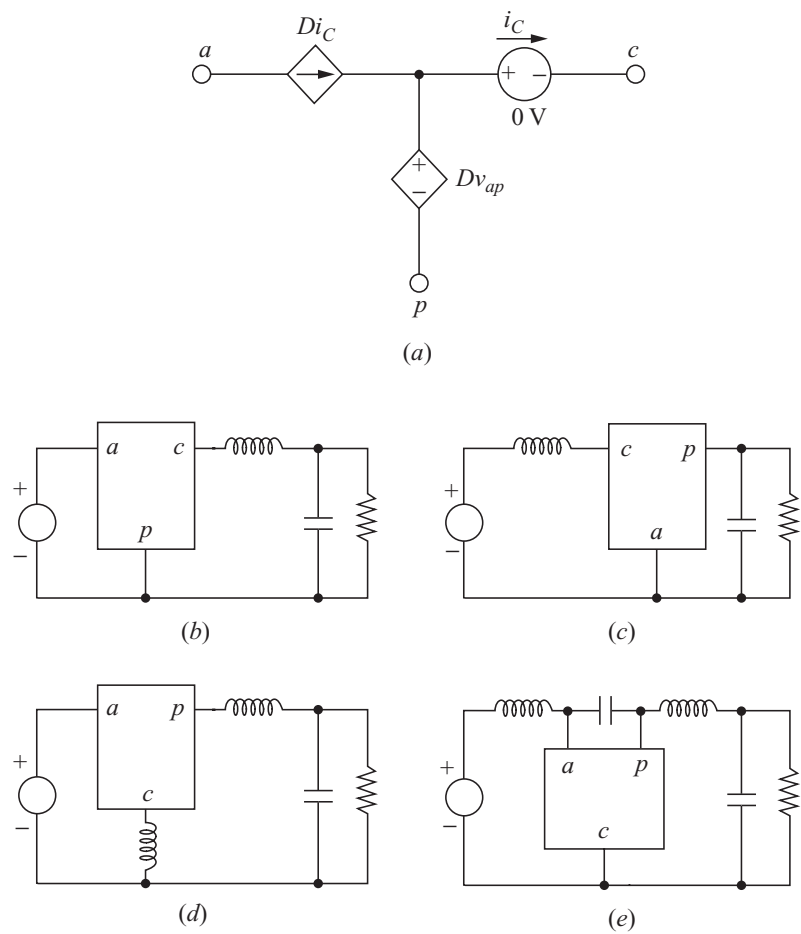


Figure 6-33
Averaged switch model in dc-dc converters. (a) PSpice averaged model for switch and diode; (b) Buck equivalent; (c) Boost equivalent; (d) Buck-boost equivalent; (e) Cuk equivalent.

The averaged switch model can be used to simulate the other dc-dc converters discussed in this chapter. Figure 6-33 shows how the average switch model is used in the boost, buck-boost, and Ćuk converters for continuous-current operation. The designation of the switch terminals a , p , and c represents active, passive, and common terminals.

6.14 Summary

- A switched-mode dc-dc converter is much more efficient than a linear converter because of reduced losses in the electronic switch.
- A buck converter has an output voltage less than the input.
- A boost converter has an output voltage greater than the input.
- Buck-boost and Ćuk converters can have output voltages greater than or less than the input, but there is a polarity reversal.
- A SEPIC (single-ended primary-inductor converter) can have an output voltage greater than or less than the input with no polarity reversal.
- Output voltage is generally reduced from the theoretical value when switch drops and inductor resistances are included in the analysis.
- Capacitor equivalent series resistance (ESR) may produce an output voltage ripple much greater than that of the capacitance alone.
- Interleaved converters have parallel switch/inductor paths to reduce the current variation in the output capacitor.
- Discontinuous-current modes for dc-dc converters are possible and sometimes desirable, but input-output relationships are different from those for the continuous-current modes.
- Switched-capacitor converters charge capacitors in one configuration and then use switches to reconnect the capacitors to produce an output voltage different from the input.
- PSpice can be used to simulate dc-dc converters by using a voltage-controlled switch or by using an averaged circuit model.

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